

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 412 — Past Questions & Materials
by Renance

Course: CSC 412
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 400

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Paper 1 of 5 — CSC 412 POSSIBLE QUESTIONS 2025

Level 400 • First Semester • PastQuestion

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Question 5

Tools for Web Development and Distributed Systems

- Define **Webpack** and state its primary function in web development. *(2 marks)*
- Describe two key features of **React** that make it suitable for building modern web applications. *(4 marks)*
- Explain one advantage and one disadvantage of distributed computing systems, focusing on reliability and complexity. *(4 marks)*
- In the context of web development tools, explain the role of **Postman** in API development and describe how it can be used to test an API. *(6 marks)*
- For a distributed system hosting a cloud-based file-sharing application, suggest one web development tool (e.g., Node.js, GitHub) that could enhance collaboration or performance, and explain how it would be applied. *(6 marks)*

Total: 22 marks

Question 4

Network Security and Mobile Computing Applications

- a. Define **Network Security** and state its primary goal. *(2 marks)*
- b. Explain the difference between **Physical Network Security** and **Technical Network Security**, providing one example for each. *(4 marks)*
- c. Describe the role of an **Intrusion Prevention System (IPS)** in network security and how it differs from antivirus software. *(4 marks)*
- d. List and explain three benefits of mobile computing for a business, focusing on productivity, communication, and business expansion. *(6 marks)*
- e. In a scenario where a company uses mobile devices for remote work, suggest one mobile computing application (e.g., GPS, Data Access) and one network security measure to protect the devices, justifying your choices. *(6 marks)*

Total: 22 marks

Question 3

Client/Server Computing and Web Applications

- a. Define **Client/Server Computing** in the context of web applications. *(2 marks)*
- b. Describe the role of **HTTP** in client/server communication and outline one step in the process of how a browser interacts with a web server. *(4 marks)*
- c. Explain one advantage and one disadvantage of client/server computing, focusing on data management and network performance. *(4 marks)*
- d. Outline the steps involved in **Building a Web Application**, specifically focusing on the "Design the Database" and "Test and Troubleshoot" stages. *(6 marks)*
- e. For an e-commerce web application, suggest one tool from the provided list (e.g., React, Postman) that could be used during development and explain how it would be applied. *(6 marks)*

Total: 22 marks

Question 2 (Compulsory)

Wireless Computing and Network Security

- a. Define **Wireless Computing** and list one example of a wireless computing technology. *(2 marks)*
- b. Explain two advantages of wireless computing over wired networks, focusing on flexibility and cost-effectiveness. *(4 marks)*
- c. Describe two disadvantages of wireless computing, specifically addressing security and performance issues. *(4 marks)*
- d. In the context of network security, explain the purpose of **Network Segmentation** and how it enhances security. *(6 marks)*
- e. Given a scenario where a university implements a wireless network for students, recommend one network security measure (e.g., Firewall, VPN) to protect the network and justify your choice. *(6 marks)*

Total: 22 marks

Question 1 (Compulsory)

Distributed Computing and Mobile Computing

In the context of net-centric computing at FUPRE:

- a. Define **Distributed Computing** and provide one example of a distributed computing system. *(2 marks)*
- b. Explain the concept of **Horizontal Scaling** in distributed computing systems and why it is considered an advantage over centralized systems. *(5 marks)*
- c. Describe the **Peer-to-Peer Architecture** in distributed computing, highlighting how it differs from a client-server model. *(5 marks)*
- d. In mobile computing, explain the role of **Communication Protocols** and provide one example of a protocol used. *(5 marks)*
- e. For a mobile banking application, suggest one type of mobile communication configuration (e.g., Mobile and Wireless) and justify why it is suitable for the application. *(5 marks)*

Total: 22 marks

Paper 2 of 5 — CSC 412 - Net-Centric Computing

Level 400 • First Semester • Handout

Lecture notes on

Net-centric Computing

Csc 412

DEPARTMENT OF COMPUTER SCIENCE,
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COURSE OUTLINE

- Introduction to Net-Centric Computing
- Distributed Computing
- Mobile and Wireless Computing
- Network Security
- Client/Server Computing (using the web)
- Building Web Applications

COURSE summary

The goal of this course is to familiarize the student with the fundamentals of computer networks, network types and categories, distributed computing, dynamic devices and mode of transmission, security of connected and communicating systems using ciphers, Client and Servers communication, and building of web applications. Upon completing this course, you would have been equipped with the knowledge of Net-centric computing fundamentals, what network is all about and wireless technologies, cloud computing and service models, and much more.

Introduction to Net-Centric Computing

For several years, major changes are being brought to the world by universal networking capabilities, such as the Internet. Today's technology solutions represent the convergence of computing power, networking capability and the data that forms the content of these solutions. At the centre these solutions net-centric computing lies.

Net-centric computing refers to an emerging technology architecture for modern distributed computing systems and applications, and an evolutionary stage of client/server computing. Its architecture is built on open standards that supports different people to collaborate and to reach different information sources in different ways. The evolutionary nature of net-centric computing links, technological capabilities and strategic opportunities, thus helping people in facing today's new problems, and providing the flexibility to meet tomorrow's challenges.

Definition: Net-Centric Computing (NCC) is a distributed environment where applications and data are downloaded from servers and exchanged with peers across a network. Net-centric Computing focuses on large-scale distributed computing systems and applications that communicate through open, wide-area networks like the Internet. General examples of large-scale network-centric systems are the World-Wide Web and Computational Grids.

Net-Centric Computing itself is the future of computing that brings together the lives of people.

The earliest computer systems functioned in isolation from one another; but today it is almost impossible to find a computer system that is not networked in some way to other systems. This includes not only desktop and laptop computers, but also devices such as cell phones, entertainment devices and this will face massive changes in the future with advancing technologies like IoT (Internet of Things) and AI (Artificial Intelligence).

OBJECTIVES OF NET-CENTRIC COMPUTING

- Net-centric computing refers to a computing paradigm that emphasizes the use of networked systems and services to enable communication, collaboration, and information sharing.
- It involves the integration of various technologies, such as computer networks, internet protocols, distributed computing, and web-based applications, to create a seamless and interconnected computing environment.
- It focuses on leveraging the power of the network to enable efficient and effective sharing of resources, data, and services. This approach allows users to access and utilize computing resources and services from anywhere, at any time, using any device with network connectivity.
- It enables the development and deployment of distributed applications and services that can be accessed and used by multiple users simultaneously. It promotes interoperability and integration among different systems and platforms, allowing for seamless communication and collaboration across organizational boundaries.

KEY CONCEPTS IN NET-CENTRIC COMPUTING

1. Service-oriented architecture (SOA): This architectural approach emphasizes the creation of modular and loosely coupled services that can be easily combined and reused to build complex applications.
2. Cloud computing: This refers to the delivery of computing resources, such as storage, processing power, and software applications, over the internet on a pay-as-you-go basis. It enables scalability, flexibility, and cost-efficiency in deploying and managing applications.
3. Web services: These are software components that expose their functionality over the internet using standard protocols, such as HTTP and XML. Web services enable interoperability and integration among different systems and platforms.
4. Big data: Net-centric computing deals with the processing and analysis of large volumes of data generated from various sources, such as social media, sensors, and transactional systems. It involves techniques for data storage, retrieval, analysis, and visualization.

WHAT ARE THE BENEFITS OF NET-CENTRIC COMPUTING?

- NCC provides the foundation for the development of innovative applications and services that leverage the power of the network to enhance productivity, efficiency, and effectiveness in various domains.
- NCC can be considered an evolution of Software as a Service (SaaS). It allows you to manage your data, applications, and infrastructure from one interface (the cloud) without

worrying about hardware or server management issues.

□ NCC combines network-based computing with its integration of various types of information technology resources (servers, storage devices, servers, computers) into centralized repositories that are served using standard Web-based protocols such as HTTP or HTTPS via a global computer communications network like the internet.

□ NCC allows organizations to focus on their core business needs without limiting themselves by software or hardware limitations imposed on their infrastructure. In other words, when an organization adopts net-centric principles, they are able to completely virtualize its IT footprint while still being able to take advantage of modern networking technologies like Local Area Networks (LANs) and Wide Area Networks (WANs).

□ NCC is a combination of IaaS, PaaS, and SaaS. What this means is that instead of buying hardware and software for your own data centre, you buy it from the cloud provider. This gives you the ability to move your data to the cloud and access it from anywhere.

□ NCC services allows you to centralize your applications with a single interface. It provides fully managed services according to user's specific requirements, which are invoked in real-time as needed rather than being provided on-demand or already provisioned for use. This concept enables multiple distributed clients to access a single entity's application in real-time.

WHAT ARE THE CAREERS RELATED TO NET-CENTRIC COMPUTING?

NCC covers a range of specialties within computing, including: computer communication, network concepts and protocols, web standards and technologies, network security, wireless and mobile computing, and distributed systems development (including web-based systems).

A wide range of careers related to net-centric computing include but are not limited to;

- Network Administrator □ Wireless Networking Engineer
- Network Engineer □ Information Security Analyst
- Security Architect □ Web Applications Developer
- Systems Analyst □ Cloud Networking Architect
- Systems designer □ Java and/or .NET developer
- Systems developer □ Mobile applications developer

cloud Computing

Cloud computing is a general term for anything that involves delivering hosted services over the internet in order to provide easy, scalable access to computing resources and IT services. Gone are the days when businesses were compelled to store or access programs, apps directly on their hard drive. Today we have cloud technology.

In simple terms, cloud computing allows you to rent instead of buy your IT. Rather than investing heavily in databases, software, and hardware, companies opt to access their compute power via the internet, or the cloud, and pay for it as they use it. These cloud services now include, but are not limited to, servers, storage, databases, networking, software, analytics, and business

intelligence.

Definition: Cloud computing refers to the use of hosted services, such as data storage, servers, databases, networking, and software over the internet. The data is stored on physical servers, which are maintained by a cloud service provider. Computer system resources, especially data storage and computing power, are available on-demand, without direct management by the user in cloud computing.

Key Characteristics

1. Provides services such as applications, storage, and infrastructure on demand.
2. Users access services remotely through the internet.
3. Services are centrally managed but delivered through distributed systems.
4. Focuses mainly on service provision rather than task sharing.

CLOUD COMPUTING ARCHITECTURE

□ Instead of storing files on a storage device or hard drive, a user can save them on cloud, making it possible to access the files from anywhere, as long as they have access to the web.

□ The services hosted on cloud can be broadly divided into infrastructure-as-a-service (IaaS), platform-as-a-service (PaaS), and software-as-a-service (SaaS). Based on the deployment model, cloud can also be classified as public, private, and hybrid cloud.

□ Further, cloud can be divided into two different layers, namely, front-end and back-end.

o The layer with which users interact is called the front-end layer. This layer enables a user to access the data that has been stored in cloud through cloud computing software.

o The layer made up of software and hardware components, i.e., the computers, servers, central servers, and databases, is the back-end layer. This layer is the primary component of cloud and is entirely responsible for storing information securely.

□ To ensure seamless connectivity between devices linked via cloud computing, the central servers use a software called middleware that acts as a bridge between the database and applications.

CORE TECHNOLOGIES ENABLING CLOUD COMPUTING

Cloud computing depends on several underlying technologies, including:

- a. Distributed Computing forms the basic architecture of cloud systems. Cloud computing systems operate as distributed systems because they rely on multiple interconnected machines.
- b. Virtualization is an essential infrastructure technology in cloud environments which allows multiple virtual machines to run on a single physical machine. This improves resource utilization and flexibility.
- c. Web 2.0 Technologies provides interfaces through which cloud services are accessed and managed. This enables interaction through web browsers, and web services serve as the primary

access point for cloud platforms.

d. Service-Oriented Architecture (SOA) defines how cloud services are designed and delivered. SOA supports different service models such as Software as a Service (SaaS), Platform as a Service (PaaS), Infrastructure as a Service (IaaS), etc.

CLASSIFICATION OF CLOUD COMPUTING

Cloud computing can either be classified based on the deployment model or the type of service. Based on the specific deployment model, we can classify cloud as public, private, and hybrid cloud. At the same time, it can be classified as infrastructure-as-a-service (IaaS), platform-as-a-service (PaaS), and software-as-a-service (SaaS) based on the service the cloud model offers.

BASED ON DEPLOYMENT

Private Cloud (also termed internal, enterprise, or corporate cloud) 3

- A private cloud is used exclusively by a single organization. It could be hosted at the organization's location or at the cloud provider's data centre.
- Computing services are offered over a private IT network for dedicated use, and services are usually managed via internal resources that is not accessible to anyone outside the organization.
- Private clouds provide a higher level of security through company firewalls and internal hosting to ensure that an organization's sensitive data is not accessible to third-party providers.
- The drawback of private cloud, however, is that the organization becomes responsible for all the management and maintenance of the data centers, which can prove to be quite resource-intensive.
- Some major private cloud vendors include Amazon Virtual Private Cloud (VPC), Hewlett Packard Enterprise (HPE), VMware, IBM Cloud, CISCO Quickstart Private Cloud, Dell EMC, Oracle Private Cloud Appliance X8, etc.

Public Cloud –

- In a public cloud, the entire computing infrastructure is located on the premises of the cloud provider, and the provider delivers services to the customer over the internet. Customers do not have to maintain their own IT and can quickly add more users or computing power as needed.
- Unlike private cloud, the services on public cloud are available to anyone who wants to use or purchase them. These services could be free or sold on-demand, where users only have to pay per usage for the CPU cycles, storage, or bandwidth they consume.
- Public clouds can help businesses save on purchasing, managing, and maintaining on-premises infrastructure since the cloud service provider is responsible for managing the system.
- They also offer scalable RAM and flexible bandwidth, making it easier for businesses to scale their storage needs.

□ Some major public cloud providers include Google Workspace, Amazon Web Services (AWS), Microsoft offerings like Microsoft 365 and Azure, as well as streaming services.

Hybrid Cloud 3

□ As the name suggests, a hybrid cloud is a combination of both public and private clouds features. The <best of both worlds= cloud model allows a shift of workloads between private and public clouds as the computing and cost requirements change.

□ Generally, hybrid cloud customers host their business-critical applications on their own servers for more security and control, and store their secondary applications at the cloud provider's location.

□ When the demand for computing and processing fluctuates, hybrid cloud allows businesses to scale their on-premises infrastructure up to the public cloud to handle the overflow while ensuring that no third-party data centers have access to their data.

□ In a hybrid cloud model, companies only pay for the resources they use temporarily instead of purchasing and maintaining resources that may not be used for an extended period. In short, a hybrid cloud offers the benefits of a public cloud without its security risks.

□ Some examples of hybrid cloud platforms include AWS Outposts, Azure Stack, Azure Arc, Azure VMware Solution, Google Anthos, Nutanix Cloud Infrastructure, Nutanix Cloud Clusters, VMware Cloud Foundation and VMware Cloud on AWS.

BASED ON SERVICE MODEL

Infrastructure as a Service (IaaS) -

□ Infrastructure as a service or IaaS is a type of cloud computing in which a service provider is responsible for providing servers, storage, and networking over a virtual interface. IaaS enables customers to access infrastructure services on an on-demand basis via the internet.

□ The key advantage is that the cloud provider hosts the infrastructure components that provide compute, storage, and network capacity so that subscribers can run their workloads in the cloud.

□ The cloud subscriber is usually responsible for installing, configuring, securing, and maintaining any software on the cloud native solutions, such as database, middleware, and application software.

Platform as a Service (PaaS) 3

□ Platform as a service or PaaS is a type of cloud computing that provides a development and deployment environment in the cloud, which allows users to develop and run applications without the complexity of building or maintaining the infrastructure.

□ PaaS provides users the advantage of accessing the developer tools they need to build and manage mobile and web applications without investing in or maintaining the underlying infrastructure. The provider hosts the infrastructure and middleware components, and the customer accesses those services via a web browser. This allows organizations to focus on the deployment and management of their applications by liberating them from the responsibility of software maintenance, planning, and resource procurement.

□ To aid productivity, PaaS solutions need to have ready-to-use programming components that allow developers to build new capabilities into their applications, including innovative technologies such as artificial intelligence (AI), chatbots, blockchain, and the Internet of Things (IoT).

□ The right PaaS offering also should include solutions for analysts, end users, and professional IT administrators, including big data analytics, content management, database management, systems management, and security.

Software as a Service (SaaS) 3

□ SaaS is a software delivery model in which the cloud provider hosts the customer's applications at the cloud provider's location. The customer accesses those applications over the internet. Rather than paying for and maintaining their own computing infrastructure, SaaS customers take advantage of subscription to the service on a pay-as-you-go basis.

□ In the SaaS model, the cloud provider manages all the hardware, middleware, application software, and security. Also referred to as hosted software or on-demand software, SaaS makes it easy for enterprises to streamline their maintenance and support.

□ Many businesses find SaaS to be the ideal solution because it enables them to get up and running quickly with the most innovative technology available. Automatic updates reduce the burden on in-house resources. Customers can scale services to support fluctuating workloads, adding more services or features they grow.

□ A modern cloud suite provides complete software for every business need, including customer experience, customer relationship management, customer service, enterprise resource planning, procurement, financial management, human capital management, talent management, payroll, supply chain management, enterprise planning, and more.

□ Popular examples of SaaS applications include Zoom, Dropbox, Google Suite, Slack, Netflix, etc.

Other Service Models

Cloud computing also supports additional service types, including:

□ Anything as a Service (XaaS)

□ Disaster Recovery as a Service (DRaaS)

□ Communications as a Service (CaaS)

□ Network as a Service (NaaS)

□ Database as a Service (DBaaS)

□ Storage as a Service (STaaS)

□ Desktop as a Service (DaaS)

□ Monitoring as a Service (MaaS)

KEY BENEFITS

The most important reason why cloud computing is growing rapidly is the various benefits it

offers.

- **Reduced costs:** Maintaining IT systems requires big outlays of capital, something that cloud helps reduce. By using the resources provided by the cloud provider, businesses avoid the need to purchase expensive infrastructure, substantially reducing their expenditure. Cloud providers work on the pay-as-you-go model, which means businesses only pay for the services they use, further reducing costs.
- **Scalability:** Cloud allows organizations to grow their users from merely a few to thousands in a very short time. Depending on the need, a business can scale their storage needs up or down, allowing organizations to be fluid.
- **Flexibility and collaboration:** Since the data on cloud can be accessed directly via the internet, it gives employees the ability to work from anywhere, anytime. Cloud gives you the freedom to set up your virtual office anywhere you are. It also allows teams to work on a project across locations by giving them access to the same files as third-party vendors.
- **Business continuity:** Cloud safely stores and protects your data in the event of an outage or crisis. This makes it easier to resume work once the systems are up and running again.
- **Competitive edge:** Cloud takes care of various business aspects, such as maintaining the IT infrastructure, licensing software, or training personnel to manage your data. It, therefore, gives you an edge over your competitors since the time and resources you invest are minimal.

CHALLENGES

Now let's move on to discuss some challenges of cloud computing.

Downtime: Almost every cloud user will tell you that outages top their list of cloud computing challenges. At times, cloud service providers may get overwhelmed due to the huge number of clients they provide services to each day. This may lead to technical outages, due to which your applications may temporarily experience some downtime.

Internet connection dependency: A user may not be able to access data on the cloud without a good internet connection and a compatible device. Moreover, using public Wi-Fi to access your files could pose a threat if the right security measures are not taken.

Financial commitment: Cloud providers use a pay-as-you-go pricing model. However, businesses need to give a monthly or annual financial commitment for most subscription plans. This needs to be factored into their operating costs.

Limited access: A user may have minimal control since the cloud service provider owns and manages the infrastructure. The user would only be able to manage applications and not the backend infrastructure. Crucial tasks, such as firmware management, may not be passed to the user at all. You always have to trust a third-party vendor to ensure security and take care of your data.

Security risks: Even if your cloud service provider promises you that they have the most reliable security certifications, there's always a chance of losing your data. With hackers increasingly targeting cloud storage to gain access to sensitive business data, this might be

an even greater concern, for which the appropriate measures need to be taken.

Distributed Computing

A distributed computer system consists of multiple software components that are on multiple computers, but run as a single system. The computers that are in a distributed system can be physically close together and connected by a local network, or they can be geographically distant and connected by a wide area network.

Definition: Distributed computing is a model in which components of a software system are shared among multiple computers or nodes. Even though the software components may be spread out across multiple computers in multiple locations, they are run as one system. This is done to improve efficiency and performance. The systems on different networked computers communicate and coordinate by sending messages back and forth to achieve a defined task.

A distributed system can consist of any number of possible configurations, such as mainframes, personal computers, workstations, minicomputers, and so on. The goal of distributed computing is to make such a network work as a single computer. Common use cases of a distributed systems are electronic banking systems, massive multiplayer online games, and sensor networks.

HOW DOES DISTRIBUTED COMPUTING WORK?

Distributed computing works by computers passing messages to each other within the distributed systems architecture.

- Each component of the system works to achieve a common goal and the end-user views results as one combined unit.
- Each component has its own end-user and the distributed system facilitates sharing resources or communication services.
- Communication protocols or rules create a dependency between the components of the distributed system. This interdependence is called coupling, and there are two main types of coupling:

1. Loose coupling 3 In loose coupling, components are weakly connected so that changes to one component do not affect the other (i.e., the components can function independently).
2. Tight coupling 3 In tight coupling, two or more components that interact with one another are dependent and must be present and functioning at the same time (i.e., every component is dependent on the other).

WHAT ARE THE TYPES OF DISTRIBUTED COMPUTING ARCHITECTURE?

In distributed computing, you design applications that can run on several computers instead of on just one computer. You achieve this by designing the software so that different computers perform different functions and communicate to develop the final solution.

There are four main types of distributed architecture:

1. Client-server architecture 3 Clients make requests to the servers for data, then formats it and displays it to the end-user. Servers manage access to resources and respond to client requests with data or status information.

2. Three-tier architecture 3 In three-tier distributed systems, client machines remain as the first tier you access. Server machines, on the other hand, are further divided into two categories:

- a. Application servers act as the middle tier for communication. They contain the application logic or the core functions that you design the distributed system for.
- b. Database servers act as the third tier to store and manage the data. They are responsible for data retrieval and data consistency.

3. N-tier architecture 3 N-tier models include several different client-server systems communicating with each other to solve the same problem. Most modern distributed systems use an n-tier architecture with different enterprise applications working together as one system behind the scenes.

4. Peer-to-peer architecture 3 Peer-to-peer distributed systems assign equal responsibilities to all networked computers. There is no separation between client and server computers, and any computer can perform all responsibilities. Peer-to-peer architecture has become popular for content sharing, file streaming, and blockchain networks.

CHARACTERISTICS OF DISTRIBUTED SYSTEMS

Computers in distributed systems may be on separate continents, in the same building, or the same room. However, they share the following:

- Heterogeneity. Usually, different platforms working together as one.
- Independent Failures. Each computer can fail independently of the others and independent from the communication channel.
- Parallel activities. Autonomous components executing concurrent tasks.
- Communication via message passing. No shared memory.
- Resource sharing. Printer, database, other services.
- No global state. No single process can have knowledge of the current global state of the system.
- No global clock. Only limited precision for processes to synchronize their clocks.

WHAT ARE THE ADVANTAGES OF DISTRIBUTED COMPUTING?

- 1. Scalability 3 Distributed computing systems can grow with your workload and requirements. You can add more computing devices (new nodes), to the distributed computing network when they are needed.
- 2. Consistency 3 Computers in a distributed system share information and duplicate data between them, but the system automatically manages data consistency across all the different computers.
- 3. Transparency 3 Distributed systems provide logical separation between the user and the physical devices. You can have different hardware, middleware, and software systems working together without worrying about the setup and configuration of individual machines.

4. Performance 3 Distributed computing can help improve performance by having each computer in a cluster handle different parts of a task simultaneously.
5. Resilience and Redundancy 3 Multiple computers can provide the same services. This way, if one machine isn't available, others can fill in for the service.
6. Cost-effectiveness 3 Distributed computing can use low-cost, off-the-shelf hardware.
7. Efficiency 3 Complex requests can be broken down into smaller pieces and distributed among different systems. This way, the request is simplified and worked on as a form of parallel computing, reducing the time needed to compute requests.
8. Distributed Applications 3 Unlike traditional applications that run on a single system, distributed applications run on multiple systems simultaneously.

WHAT ARE SOME DISTRIBUTED COMPUTING USE CASES?

Distributed computing is everywhere today. Mobile and web applications are examples of distributed computing because several machines work together in the backend for the application to give you the correct information. However, when distributed systems are scaled up, they can solve more complex challenges.

- Healthcare and life sciences use distributed computing to model and simulate complex life science data. Image analysis, medical drug research, and gene structure analysis all become faster with distributed systems.
- In Engineering research, engineers can simulate complex physics and mechanics concepts on distributed systems. They use this research to improve product design, build complex structures, and design faster vehicles.
- Financial services firms use distributed systems to perform high-speed economic simulations that assess portfolio risks, predict market movements, and support financial decision-making.
- Energy companies need to analyse large volumes of data to improve operations and transition to sustainable and climate-friendly solutions. They use distributed systems to analyse high-volume data streams from a vast network of sensors and other intelligent devices.

CLOUD COMPUTING VS. DISTRIBUTED COMPUTING

Basis Cloud Computing Distributed Computing

Meaning Provides computing services Multiple computers share tasks to (software, storage, infrastructure) over the internet.

Main Focus Service delivery to users or Task sharing and problem solving. organizations.

Purpose Provides on-demand computing Coordinates multiple machines to resources. perform the same task.

User Access Accessed remotely via the internet. Usually operates behind the system; users may not directly interact.

Relationship Built on distributed systems. Serves as the foundation for cloud computing.

Example Google Drive, AWS, Microsoft Distributed databases, distributed Azure. processing systems.

PARALLEL COMPUTING VS. DISTRIBUTED COMPUTING

Parallel computing is a computing approach in which multiple calculations or processes are executed simultaneously using one or more computers.

Characteristics

1. Processes run concurrently.
2. Processors typically share a common memory space.
3. High-speed communication between processors.
4. Strong coordination between computing units.

Basis Parallel Computing Distributed Computing

Meaning Multiple processors execute tasks Multiple independent computers simultaneously. perform parts of a task.

Memory Structure Shared memory among processors. Each processor has its own private memory.

Coupling Level Tightly coupled system. Loosely coupled system.

Communication Through shared memory. Through message passing.

Method

Location Usually within one computer or Computers may be closely connected system. geographically separated.

Goal Increase computation speed. Solve complex problems through collaboration.

GRID COMPUTING VS. DISTRIBUTED COMPUTING

Grid computing is a distributed computing model that uses a network of computers working together to solve complex problems.

Key Characteristics

1. Multiple computers form a distributed architecture.
2. Systems perform independent tasks.
3. Machines are loosely connected through the internet or low-speed networks.
4. Emphasizes large-scale resource sharing and coordination.

Features of Grid Computing

1. Highly scalable computing model.
2. Focuses on performance and coordination across different networks.
3. Internally, each grid operates as a tightly integrated system.
4. Externally, different grids communicate loosely.
5. Each grid performs specific functions and shares results with other grids.

Basis Grid Computing Distributed Computing

Meaning Network of distributed computers Multiple computers working solving large-scale complex problems. together on a common task.

Scale Very large-scale and highly scalable. Can be small or large scale.

Network Loosely connected via internet or low- May use various connection

Connection speed networks. types.

Task Execution Independent tasks handled by different Tasks divided and coordinated nodes. among systems.

Focus Resource sharing across multiple General task distribution and networks for performance. coordination.

Structure Internally tightly coupled, externally General distributed architecture. loosely coupled.

Mobile & Wireless computing

Mobile Computing is a technology that allows transmission of data, voice and video via a computer or any other wireless enabled device without having to be connected to a fixed physical link.

Definition: Mobile computing refers to the set of IT technologies, products, services and operational strategies and procedures that enable end users to access computation, information and related resources and capabilities while mobile. Mobile most commonly refers to access in motion, where the user is not restricted to a given geographic location.

Today, mobile computing technology is everywhere. It has applications in business and consumer markets, industrial and entertainment industries, as well as many specialized vertical markets.

Desktop computers offer more hardware configuration capabilities and computational performance. However, the majority of end users prefer mobile devices. The key advantage of mobile computing is convenience, where users have access to information and computational resources anytime and anywhere.

HOW DOES MOBILE COMPUTING WORK?

□ Generally, a mobile computing system involves:

o A mobile device, such as a laptop computer, tablet or smartphone, and

o A wireless network connection based on Wi-Fi or cellular wireless technology, such as 5G.

□ Mobile devices typically can store data locally, and access to that data doesn't require a network connection.

□ Mobile computers typically accommodate access to both wireless and wired technology. Access to shared network resources, including Mobile cloud-based resources, is essential given the collaborative nature of work today.

□ Integrated, rechargeable batteries power mobile devices, and most can run on an alternating current (AC) power source when used from a fixed location.

□ Mobile device selection is based on the intended application selected for use. For example,

laptops are better suited for content creation, and tablets are often preferred for content consumption. Smartphones function as pocket-size computers and communication devices, but they have small screen sizes and screen-based keyboards.

WHY IS MOBILE COMPUTING USED?

Mobile computing is used in most facets of life both in business and by consumers.

- Users can use mobile devices like laptops, smartphones, wearables and tablets for an array of activities, including communication, entertainment, banking, bill paying and health and fitness tracking.

- It enables users to be untethered from a power source for periods of time. This is advantageous for traveling workers who want to stay connected to their work while on the move but may not have all the connectivity and power options they have in an office setting.

- It's also useful for remote workers. Employees use laptops or tablets to work and collaborate. Delivery personnel use mobile devices for logistical and management information, and to verify that deliveries were made to the right place. Field service technicians use mobile devices for field service management, tracking and support capabilities.

- Mobile devices and apps can collect user data in various environments and contexts. Fitbits and smartwatches are examples of wearable technology that collect user data in novel contexts, including fitness and health settings.

- Mobile computing also makes the internet of things (IoT) possible. Consumer and enterprise IoT devices are used to communicate with other devices without human intervention. For example, self-driving cars use sensors, onboard computers and other connected devices to connect with GPS, weather and other systems to navigate the road safely. Smart sensors are also used in supply chain management systems to track the progress of goods in transit.

WHAT ARE THE TYPES OF MOBILE COMPUTING?

Mobile computing is a combination of infrastructure, hardware and software technology. The various parts of mobile computing are as follows:

1. **Mobile Infrastructure.** Infrastructure refers to the technical pieces that allow devices to communicate. Mobile infrastructure includes the wireless networks, wireless protocols and data formats.
2. **Mobile Hardware.** The physical mobile device and supporting hardware that users interact with make up the mobile hardware. This may include cell phones, laptops, tablets, wearable computers and accompanying chargers and accessories.
3. **Mobile Software.** This refers to the applications that run on mobile devices, including mobile operating systems (OSes) and user-facing applications, such as mobile browsers and e-commerce applications.

MOBILE CLASSIFICATION

Mobile computing is not only limited to mobile phones, but there are various gadgets available in the market that are built on a platform to support mobile computing. They are usually classified in the following categories:

□ Personal Digital Assistant (PDA)

□ The main purpose of this device is to act as an electronic organizer or day planner that is portable, easy to use and capable of sharing information with your computer systems.

□ PDA is an extension of the PC, not a replacement. These systems are capable of sharing information with a computer system through a process or service known as synchronization. Both devices will access each other to check for changes or updates in the individual devices. The use of infrared and Bluetooth connections enables these devices to always be synchronized.

□ With PDA devices, a user can browse the internet, listen to audio clips, watch video clips, edit and modify office documents, and many more services. The device has a stylus and a touch sensitive screen for input and output purposes.

□ Smartphones

□ It combines the features of a PDA with that of a mobile phone or camera phone. It has a superior edge over other kinds of mobile phones.

□ Smartphones have the capability to run multiple programs concurrently. These phones include high-resolution touch screens, and web browsers that can access and properly display standard web pages rather than just mobile-optimized sites, and can access high-speed data via Wi-Fi and high-speed cellular broadband.

□ The most common mobile Operating Systems (OS) used by modern smartphones include, Google's Android, Apple's iOS, Nokia's Symbian, RIM's BlackBerry OS, Samsung's Bada, Microsoft's Windows Phone, and embedded Linux distributions such as Maemo and MeeGo.

□ Such operating systems can be installed on different phone models, and typically each device can receive multiple OS software updates over its lifetime.

□ Tablet PC and iPads

□ This mobile device is larger than a mobile phone or a PDA. It integrates into a touch screen and is operated using touch sensitive motions on the screen. They are often controlled by a pen or by the touch of a finger.

□ They are usually in slate form and are light in weight. Examples would include iPads, Galaxy Tabs, Blackberry Playbooks etc.

□ They offer the same functionality as portable computers and they support mobile computing in a far superior way and have enormous processing horsepower.

□ Users can edit and modify document files, access high speed internet, stream video and audio data, receive and send e-mails, attend/give lectures and presentations among its very many other functions. They have excellent screen resolution and clarity

WHAT ARE THE ADVANTAGES OF MOBILE COMPUTING?

1. Flexibility 4 Mobile computing allows you to work from anywhere, at any time, giving you the flexibility to get things done when and where it's convenient for you.
2. Connectivity 4 Mobile computing keeps you connected to the Internet, allowing you to access information, stay in touch with others, and collaborate with colleagues and classmates.
3. Efficiency 4 Mobile computing helps you be more efficient, allowing you to complete tasks quickly and easily while on the go.
4. Productivity 4 Mobile computing can increase productivity, as you can take care of important tasks and responsibilities while away from your desk.
5. Innovation 4 Mobile computing is constantly advancing and evolving, providing new and innovative ways to stay connected and get things done.

WHAT ARE THE LIMITATIONS OF MOBILE COMPUTING?

1. Connectivity 4 While the mobile infrastructure continues to improve, there are areas where signal strength is poor or non-existent.
2. Distraction 4 Mobile devices can be distracting and potentially dangerous in a hazardous work environment that requires the employee's attention, such as a construction site. They pose dangers if used inappropriately while driving.
3. Dependence 4 Mobile computing can lead to a dependence on technology, making it difficult to disconnect and enjoy life without constant digital distraction.
4. Battery Life 4 Mobile devices have limited battery life, which requires frequent recharging, thus causing inconvenience and disruption.
5. Security Concerns 4 Mobile devices can be vulnerable to cyberattacks and data theft, putting personal information and sensitive data at risk.
6. Cost 3 Mobile computing can be expensive, with high costs for devices, data plans, and accessories.
7. Physical Damage 4 Mobile devices are small and fragile, making them susceptible to physical damage from drops, spills, and other accidents.

MOBILE COMPUTING SECURITY THREATS

Mobile computing has its fair share of security concerns as any other technology. The first step to securing mobile devices is knowing the risks that are lurking on the Internet that might be targeted at you. Cybercriminals can be after your data, your credentials, or your money, and they can use multiple ways to get what they want, some of the which includes:-

□ Phishing 3 This is a scamming attempt to steal users9 credentials or sensitive data, such as

credit card numbers. Fraudsters send users emails or short message service (SMS) messages (commonly known as text messages) designed to look as though they're coming from a legitimate source, using fake hyperlinks.

□ **Malware and Ransomware** 3 Mobile malware is undetected software, such as a malicious app or spyware, created to damage, disrupt or gain illegitimate access to a client, computer, server or computer network. Ransomware, a form of malware, threatens to destroy or withhold a victim's data or files unless a ransom is paid to decrypt files and restore access.

□ **Cryptojacking** 3 This is a form of malware, uses an organization's computing power or individual's computer power without their knowledge to mine cryptocurrencies such as Bitcoin or Ethereum, decreasing a device's processing abilities and effectiveness.

□ **Unsecured Wi-Fi** 3 Hotspots without a virtual private network (VPN) make mobile devices more vulnerable to cyberattack. Cybercriminals can intercept traffic and steal private information using methods such as man-in-the-middle (MitM) attacks. Cybercriminals can also deceive users into connecting to rogue hotspots, making it easier to extract corporate or personal data.

□ **Outdated operating systems** usually contain vulnerabilities that have been exploited by cybercriminals, and devices with outdated OS remain vulnerable to attack. Manufacturer updates often include critical security patches to address vulnerabilities that may be actively exploited.

□ **Excessive app permissions.** Mobile apps have the power to compromise data privacy through excessive app permissions. App permissions determine an app's functionality and access to a user's device and features, such as its microphone and camera. Some apps are riskier than others. Some can be compromised, and sensitive data can be funnelled through to untrustworthy third parties.

□ **The Internet of Things (IoT).** Over the years, the number and type of devices 3 from phones to smartwatches and wireless appliances 3 that can access your network has grown so fast, which can be hard to monitor and secure with an antivirus solution. That is why IoT devices can be tempting for hackers who target them as an entry point to your home or business network.

□ **Password security.** Bad password habits like reusing passwords or choosing weak passwords from the get-go just make a cybercriminal's work much easier. Additionally, you can be at risk for credential-based brute-force cyberattacks like credential stuffing or password spraying.

NETWORK SECURITY

The transmission of data from one point (A) on the network to the other point (B) is a great concern and therefore, there is the need to deploy measures that can secure the transmission of data away from unauthorized individuals. Hence, the need for network security.

Definition

□ Network security is a term that describes the security tools, tactics and security policies

designed to monitor, prevent and respond to unauthorized network intrusion, while also protecting digital assets, including network traffic.

□ Network security includes hardware and software technologies (including resources such as savvy security analysts, hunters, and incident responders) and is designed to respond to the full range of potential threats targeting your network.

□ Network security is the protection of the underlying networking infrastructure from unauthorized access, misuse, or theft. It involves creating a secure infrastructure for devices, applications, users, and applications to work in a secure manner.

Data as the Life-Blood of Businesses

Data is the lifeblood of any business. It supports business growth, carries vital resources and helps organizations to stay healthy. And if data is blood, then the network is the heart beat that pumps data through the business system. But modern cyber threats are like vampires, doing everything possible to get that blood that keeps any business going. In order to defend against these vampiric threats and save your business from data loss that could potentially be worth millions, you need more than just a stake and some cloves of garlic; you need effective, robust network security and network visibility.

How does network security work?

Network security combines multiple layers of defenses at the edge and in the network. Each network security layer implements policies and controls. Authorized users gain access to network resources, but malicious actors are blocked from carrying out exploits and threats.

Key Focuses of Network Security

There are three key focuses that should serve as a foundation of any network security strategy: protection, detection and response.

□ Protection entails any tools or policies designed to prevent network security intrusion.

□ Detection refers to the resources that allow you to analyse network traffic and quickly identify problems before they can do harm.

□ Response is the ability to react to identified network security threats and resolve them as quickly as possible.

Network Security Tools and Techniques

Enterprises9 network face threats of all shapes and sizes, and thus should be prepared to defend, identify and respond to a full range of attacks. But the reality is that the biggest danger to most companies are not fly-by-night threat actors, but rather attackers that are well-funded and are targeting specific organizations for specific reasons. For that reason, your network security strategy needs to be able to address the various methods these actors might employ.

Here are some network security tools and techniques designed to help you do just that:

Access control

If threat actors cannot access your network, the amount of damage they will be able to do will be extremely limited. But in addition to preventing unauthorized access, be aware that even authorized users can also be potential threats. Access control allows you to increase your network security by limiting user access and resources to only the parts of the network that directly apply

to individual users' responsibilities.

Anti-malware software

Malware, in the form of viruses, trojans, worms, keyloggers, spyware, etc. are designed to spread through computer systems and infect networks. Anti-malware tools are a kind of network security software designed to identify dangerous programs and prevent them from spreading. Anti-malware and antivirus software may also be able to help resolve malware infections, minimizing the damage to the network.

Anomaly detection

It can be difficult to identify anomalies in your network without a baseline understanding of how that network should be operating. Network anomaly detection engines (ADE) allow you to analyse your network, so that when breaches occur, you'll be alerted to them quickly enough to be able to respond.

Application security

For many attackers, applications are a defensive vulnerability that can be exploited. Application security helps establish security parameters for any applications that may be relevant to your network security.

Data loss prevention (DLP)

Often, the weakest link in network security is the human element. DLP technologies and policies help protect staff and other users from misusing and possibly compromising sensitive data or allowing said data out of the network.

Email security

As with DLP, email security is focused on shoring up human-related security weaknesses. Via phishing strategies (which are often very complex and convincing), attackers persuade email recipients to share sensitive information via desktop or mobile device, or inadvertently download malware into the targeted network. Email security helps identify dangerous emails and can also be used to block attacks and prevent the sharing of vital data.

Endpoint security

The business world is becoming increasingly bring your own device (BYOD), to the point where the distinction between personal and business computer devices is almost non-existent. Unfortunately, sometimes the personal devices become targets when users rely on them to access business networks. Endpoint security adds a layer of defense between remote devices and business networks.

Firewall

Firewall function much like gates that can be used to secure the borders between your network and the internet. Firewalls are used to manage network traffic, allowing authorized traffic through while blocking access to non-authorized traffic.

Intrusion prevention systems

Intrusion prevention systems (also called intrusion detection) constantly scan and analyse network traffic/packets, so that different types of attacks can be identified and responded to quickly. These systems often keep a database of known attack methods, so as to be able to

recognize threats immediately.

Network segmentation

There are many kinds of network traffic, each associated with different security risks. Network segmentation allows you to grant the right access to the right traffic, while restricting traffic from suspicious sources.

Security information and event management (SIEM)

Sometimes simply pulling together the right information from so many different tools and resources can be prohibitively difficult 4 particularly when time is an issue. SIEM tools and software give responders the data they need to act quickly.

Virtual private network (VPN)

VPN tools are used to authenticate communication between secure networks and an endpoint device. Remote-access VPNs generally use IPsec or Secure Sockets Layer (SSL) for authentication, creating an encrypted line to block other parties from eavesdropping.

Remote Access VPN

Remote access VPN provides remote and secure access to a company network to individual hosts or clients, such as telecommuters, mobile users, and extranet consumers. Each host typically has VPN client software loaded or uses a web-based client. Privacy and integrity of sensitive information is ensured through multi-factor authentication, endpoint compliance scanning, and encryption of all transmitted data.

Zero Trust Network Access (ZTNA)

The zero trust security model states that a user should only have the access and permissions that they require to fulfil their role. This is a very different approach from that provided by traditional security solutions, like VPNs, that grant a user full access to the target network. Zero trust network access (ZTNA) also known as software-defined perimeter (SDP) solutions permits granular access to an organization's applications from users who require that access to perform their duties.

Web security

Including tools, hardware, policies and more, web security is a blanket term to describe the network security measures businesses take to ensure safe web use when connected to an internal network. This helps prevent web-based threats from using browsers as access points to get into the network.

Wireless security

Generally speaking, wireless networks are less secure than traditional networks. Thus, strict wireless security measures are necessary to ensure that threat actors aren't gaining access.

Sandboxing

Sandboxing is a cybersecurity practice where you run code or open files in a safe, isolated environment on a host machine that mimics end-user operating environments. Sandboxing observes the files or code as they are opened and looks for malicious behavior to prevent threats from getting on the network. For example malware in files such as PDF, Microsoft Word, Excel

and PowerPoint can be safely detected and blocked before the files reach an unsuspecting end user.

Hyperscale Network Security

Hyperscale is the ability of an architecture to scale appropriately, as increased demand is added to the system. This solution includes rapid deployment and scaling up or down to meet changes in network security demands. By tightly integrating networking and compute resources in a software-defined system, it is possible to fully utilize all hardware resources available in a clustering solution.

Cloud Network Security

Applications and workloads are no longer exclusively hosted on-premises in a local data center. Protecting the modern data centre requires greater flexibility and innovation to keep pace with the migration of application workloads to the cloud. Software-defined Networking (SDN) and Software-defined Wide Area Network (SD-WAN) solutions enable network security solutions in private, public, hybrid and cloud-hosted Firewall-as-a-Service (FWaaS) deployments.

ROBUST NETWORK SECURITY WILL PROTECT AGAINST

Virus

A virus is a malicious, downloadable file that can lay dormant that replicates itself by changing other computer programs with its own code. Once it spreads those files are infected and can spread from one computer to another, and/or corrupt or destroy network data.

Worms

Can slow down computer networks by eating up bandwidth as well as the slow the efficiency of your computer to process data. A worm is a standalone malware that can propagate and work independently of other files, where a virus needs a host program to spread.

Trojan

A trojan is a backdoor program that creates an entryway for malicious users to access the computer system by using what looks like a real program, but quickly turns out to be harmful. A trojan virus can delete files, activate other malware hidden on your computer network, such as a virus and steal valuable data.

Spyware

Much like its name, spyware is a computer virus that gathers information about a person or organization without their express knowledge and may send the information gathered to a third party without the consumer's consent.

Adware

Can redirect your search requests to advertising websites and collect marketing data about you in the process so that customized advertisements will be displayed based on your search and buying history.

Ransomware

This is a type of trojan cyberware that is designed to gain money from the person or organization's

computer on which it is installed by encrypting data so that it is unusable, blocking access to the user's system.

Benefits of Network Security

Network Security Builds trust

Security for large systems translates to security for everyone. Network security boosts client and consumer confidence, and it protects your business from the reputational and legal fallout of a security breach.

Network Security Mitigates risk

The right network security solution will help your business stay compliant with business and government regulations, and it will minimize the business and financial impact of a breach if it does occur.

Network Security Protects proprietary information

Your clients and customers rely on you to protect their sensitive information. Your business relies on that same protection, too. Network security ensures the protection of information and data shared across the network.

Network Security Enables a more modern workplace

From allowing employees to work securely from any location using VPN to encouraging collaboration with secure network access, network security provides options to enable the future of work. Effective network security also provides many levels of security to scale with your growing business.

Client/Server Computing

(using the web)

In client-server computing, the clients request a resource and the server provides that resource. A server may serve multiple clients at the same time while a client is in contact with only one server. Both the client and server usually communicate via a computer network but sometimes they may reside in the same system.

Characteristics of Client Server Computing

The salient points for client server computing are as follows:

- The client server computing works with a system of request and response. The client sends a request to the server and the server responds with the desired information.
- The client and server should follow a common communication protocol so they can easily interact with each other. All the communication protocols are available at the application layer.
- A server can only accommodate a limited number of client requests at a time. So, it uses a system based on priority to respond to the requests.
- Denial of Service (DoS) attacks hinder a server's ability to respond to authentic client requests by inundating it with false requests.
- An example of a client server computing system is a web server. It returns the web pages to the clients that requested them.

Difference between Client Server and Peer-to-Peer Computing

The major differences between client server computing and peer to peer computing are as follows:

- In client server computing, a server is a central node that services many client nodes. On the other hand, in a peer-to-peer system, the nodes collectively use their resources and communicate with each other.
- In client server computing the server is the one that communicates with the other nodes. In peer to peer to computing, all the nodes are equal and share data with each other directly.
- Client Server computing is believed to be a subcategory of the peer to peer computing.

Advantages of Client Server Computing

The different advantages of client server computing are –

- All the required data is concentrated in a single place i.e., the server. So, it is easy to protect the data and provide authorisation and authentication.
- The server need not be located physically close to the clients. Yet the data can be accessed efficiently.
- It is easy to replace, upgrade or relocate the nodes in the client server model because all the nodes are independent and request data only from the server.
- All the nodes (clients and server) may not be built on similar platforms yet they can easily facilitate the transfer of data.

Disadvantages of Client Server Computing

The different disadvantages of client server computing are –

- If all the clients simultaneously request data from the server, it may get overloaded. This may lead to congestion in the network.
- If the server fails for any reason, then none of the requests of the clients can be fulfilled. This leads of failure of the client server network.
- The cost of setting and maintaining a client server model are quite high.

Building Web Applications

A Web app is an interactive computer program, built with web technologies (HTML, CSS, JS), which stores (Database, Files) and manipulates data (CRUD), and is used by a team or single user to perform tasks over the internet. The HTML and the CSS serves as the front-end to receive data from the user while the database, programming like JavaScript and PHP serves as the back-end.

Prerequisites for Building a Web Application

To make a data-centric web app from the bottom-up, it is advantageous to understand:

1. Backend language (e.g., Python, Ruby) - control how your web app works
2. Web front end (HTML, CSS, JavaScript) - for the look and feel of your web app
3. DevOps (GitHub, Jenkins) - Deploying / hosting your web app

If you do not have any experience with the points above, you need not worry. You have two options:

- Learn the points above. There are lots of resources online to help you. I'd recommend:
 - o FreeCodeCamp
 - o Codecademy
 - o Codewars
- Use a web app builder. Web app builders allows you to build web apps without writing any code. I'd recommend:
 - o Google AppSheet
 - o Bubble
 - o Glide

Building a Web Application

Step 1 – Source an Idea

Before making a web app, you must first understand what you intend on building, and more importantly, the reason for it. Your idea should stem from solving someone's problem. Ideally, it is important that you choose an idea which is of your interest. This can be done by asking eye questions like:

- How much time do I have to build this app?
- What am I interested in?
- What apps do I enjoy using?
- What do I like about these apps?
- How much time/money will this app save or generate for me (as a user)?
- How much will it improve my life?

Step 2 – Market Research

Once you have chosen your idea(s), it is important to research the market to see if a similar product exists, and if a market exists. The number major reason why start-ups fail is due to the failure of to achieving product-market fit. Product/market fit means being in a good market with a product that can satisfy that market. To quickly find out if a similar web app exists, use the following tools to search for your idea:

- Google
- Patent and trademark search
- Betalist

If a similar product exists, do not worry. This can be a sign that a market for your new idea exists. Your future competitors have laid the groundwork, educated the market. It's time for you to swoop in and steal the thunder.

If a similar product does not exist, it is a possibility you have struck lucky. On the other hand, it is a possibility someone before has ventured down this path and hit a dead-end. Nobody wants to experience that, so it is important to dive deep into the market and source the wisdom of:

1. Your Web App's target market - Share your web app idea on forums related to your target market. If you know anyone who works within your target market, explain your idea to them. The more you talk and receive validation from your target market, the better.
2. Google Trends - A quick search of your web app idea will reveal relating trends.

3. SEO tool 3 Write a list of keywords relating to your web app. If it's an 8OKR tool, use the tools to search 8OKR tool, 8OKR app, and 8objectives and key results software. If the SEO tool indicates there are lots of people searching for your keyword terms, this is a small indicator you have a target market.

4. Social Media - Jump over to Twitter/Facebook groups and present your idea to your target market.

5. Events - If there is a local event in your area attracting people from your target market, go to it. Share your idea and record the feedback. After completing the above steps, you should have enough information to understand if there's a market for your product. If there is a market for your product, and there's also established competition, it's important to research them.

Step 3 - Define your web apps functionality

You have got your idea, you have validated the market, it is now time to list everything you want your app to do. A common mistake here is to get carried away. The more functionality you add, the longer it will take to build your web app. Quite often, the longer a web app takes to build, the more frustration you will experience.

Only define functionality which solves your target markets problems. Remember, your web app is a work in progress and the first goal is version 1. It will still have cool features and delight your users, but you must keep things simple. For direction, below is a list of basic functions required for a simple Customer Relationship Management (CRM) app:

- ☐ Users can create an account
- ☐ Users can retrieve lost passwords
- ☐ Users can change their passwords
- ☐ Users can create new contacts
- ☐ Users can upload new contacts
- ☐ Users can assign a value to contacts
- ☐ Users can write notes under contacts
- ☐ Users can label a contact as a lead, customer, or associate
- ☐ Users can filter contacts by lead, customer, or associate
- ☐ Users can view the total value of leads, customers and associates

The above list will help you define your features. Once you're done, roll up your sleeves. It's time to get creative! Moving from the Ideation stage, to design stage.

Step 4 - Sketch your web app

There are multiple stages of designing a web app. The first stage is sketching using a notebook (with no lines) and pen/pencil.

After step 1, 2 and 3, you should have an idea of what your web app is, who your users are, and the features it will have. Sketch out the wireframe of your web apps UI (User Interface). It does not have to be exact; it is just a sketch. When sketching, consider the following:

- ☐ Navigation
- ☐ Branding

- Forms
- Buttons
- Any other interactive elements

Sketch different versions of your web app. Consider how your web app's functionality will affect the overall design. Annotate your sketch and outline how your app should work. Taking notes will help you clarify and understand why you have designed certain elements at a later stage. Overcomplicating the design at this stage will only lead to frustration.

Step 5 – Plan your web apps workflow

It is time to put yourself in the shoes of your user. Here, you plan your web apps workflow. First, go back to step 2 and look at your market research. Take your list of competitors and sign up to their free trials. Have a quick play around with their product. Write notes on what you thought was good and what you thought was bad. Pay particular attention to the workflow.

After you have finished analysing your competitor's web apps, it is time to write down different workflows for your app. Consider the following points:

- How does a user signup?
- Do they receive a verification email?
- How does a user login?
- How does a user change their password?
- How does a user navigate through the app?
- How does a user change their user settings?
- How does a user pay for the app?
- How does a user cancel their subscription?

All of a sudden our one-page web app turns into a 10-page web app. Write a list of all the different pages your web application will have. Consider the different states of pages. For example, the homepage will have two states; logged in and logged out. Logged in users will see a different page than logged out users, and so on.

Step 6 – Wireframing / Prototyping Your Web Application

Ok, it's time to turn those sketches and that new-found understanding of your web application into a wireframe/prototype. Wireframing is the process of designing a blueprint of your web application. Prototyping is taking wireframing a step further, adding an interactive display.

The decision is to wireframe or prototype is up to you. If you have the time, I would have recommended prototyping as it will make it easier to communicate your web app when seeking validation. You can prototype/wireframe using the following tools:

Sketch (macOS)

InVision (macOS)

Adobe XD (macOS, Windows)

Figma (Web, macOS, Windows, Linux)

Balsamiq (macOS, Windows, Web)

I recommend you create a design system/style guide first. You can find inspiration at UXPin. Design systems improve design consistency. But it's not required.

Step 7 – Seek early validation

You have now got a beautiful wireframe/prototype which visually describes your web app. It is time to show your beautiful wireframe to the world. At this stage we want constructive feedback.

Simply asking your friends would they use your new web app is not enough. You should start with a small number of representative users. Go to your target market's forums, watering holes, their places of work and verify the problem with them, and present your solution. Try to build a rapport with these representatives as they could become your customers. Once you have gathered feedback and product validation about your prototype, you can start building your web app.

However, before Starting the development stage, take note of the following tips:

- Attempt to get a small section of your app fully working. Building the smallest possible section will allow you to piece all the bits together, and iron out those creases early.
- At the start - expect things to change a lot as you learn and discover what you have not thought about. Have faith that your app will stabilise and don't be afraid of making big changes.
- Spend time learning your tools. You may feel like you are wasting your time, reading, or experimenting with <hello world>. However, learning the correct way to do things will have a huge positive, cumulative effect on your productivity over time.
- Do not avoid issues that need to be fixed. Face your issues head on - they will never go away and will only grow in stature. However, If changes are still expected while building your web app, then its best to spend as little time as possible on the issues (it is a tricky balance).

Step 8 – Architect and build your database

So far, we know roughly our web application's functionality, what it looks like, and the pages required. Now it is time to determine what information we will store in our database.

A database is simply a collection of data! Data can be stored to disk, or in memory on a server, or both. You could create a folder on your hard drive, store a few documents, and call it a database.

A Database Management System (DBMS) is a system that provides you with consistent APIs to (most commonly):

- Create databases, update and delete databases,
- Read and write data to databases,
- Secure access to a database by providing levelled access to different areas and functions.

The type of data you need to store and what your users will use the data to do, will determine the type of database required to run your web app.

There are many types of database system for many different purposes. A web app will most commonly use one of the following:

1. SQL 3 You should use a SQL database if your data is very relational. Your data is relational if you have multiple, well defined record types that have relationships between them. For example, a <Customer> may have many <Invoices> stored against their record. Typically, you would create a Customer table and an Invoice table - which could be linked together

by a <Foreign Key= columns. E.g., Customer.Id = Invoice.CustomerId.

SQL databases have an extremely powerful query language that allows you to present your data in all sorts of useful ways. They have been around for decades, are very well understood, and usually a safe choice. MySQL, Postgresql, Microsoft SQLServer are some of the most common, along with many more modern offerings. The downside of SQL databases is that you must declare all your tables and columns up front.

2. Document Database 3 You should use a document database if your data is not very relational. Document databases store <documents=. Each record in your database is simply a big blob of structured data 3 often in JSON format. If you need to store relationships between your records, you will have to write code to manage this yourself. However, many other aspects of using document databases are much simpler. Your database can be <schemaless= - meaning that you do not have to declare your records9 definitions up front.

Generally speaking, document database tends to be much more scalable than SQL databases. They usually offer some querying capabilities, although sometimes not as powerful as SQL. Examples of document databases are: MongoDB, CouchDb, Firebase (serverless), Dynamo Db (AWS).

Step 9 - Build the frontend

Note: In reality, you will build your backend and frontend at the same time. But we9ll keep it simple for better understanding.

The Frontend is the visual element of your web application. It defines what you see and interact with. The frontend is developed with HTML, CSS, and JavaScript.

First, you need to set up your development environment. The components of this will be:

- A code editor, such as VS Code, Sublime Text
- A compilation, and packaging framework:
 - o Webpack,
 - o Gulp,
 - o Grunt.
- A frontend framework (strictly not necessary, but highly advised unless you are an experienced frontend developer):
 - o React,
 - o Ember,
 - o Vue,
 - o Svelte, (The list is endless!).
- Configuring your packaging tool to talk to your backend - which is most likely running on a different port on localhost. Usually, this is done using Node9s HTTP proxy. Most packaging solutions have this option built-in, or available as plugins.

Step 10 - Build your backend

The backend is typically what manages your data. This refers to databases, servers, and everything the user can9t see within a web application. Building your backend is one of the

toughest parts of web app development.

The primary job of the backend is to:

1. Provide HTTP endpoints for your frontend, which allow it to operate on your data. E.g., Create, Read, Update and Delete (<CRUD=) records.
2. Authenticate users (verify they are who they say they are before logging them in).
3. Authorization. When a logged in user makes a request, the backend will determine whether they are allowed (authorized) to perform the requested action.
4. Serve the frontend.

When building your web app, you need to choose between Single Page Application and Multiple Page Application (Server Pages). Your choice will affect how you develop your backend.

□ If you have chosen Server Pages, your backend will also be responsible for generating your frontend and serving it to your users. Server pages need their own framework:

Django

Express

Flask

□ With a single page app, the backend will simply serve your static frontend files (i.e., your <Single Page= and its related assets).

When choosing your backend, go with what's familiar.

□ Login/User & Session Management. How will users authenticate?

o Username and password?

o Open ID (i.e., sign in as Google, FB, etc)

□ What is the security best practices employed in your web app?

I highly recommend OWASP.

□ What user levels will you create in the system?

□ Creating multiple environments is highly recommended. For example:

o Testing environment is needed for all the latest development features.

o Beta environment is needed to give early releases to clients.

o Production environment is needed for your live web app system.

Step 11 - Host your web application

Hosting involves running your web app on a particular server. To host your web app, follow these quick steps:

1. Buy a domain (e.g., Namecheap).
2. Buy/Setup an SSL certificate (for encryption).
3. Choose a cloud provider:

Amazon, Vercel,

MS Azure, Heroku,

Google Cloud Platform, Firebase, etc..

DigitalOcean,

Alternatively, the above steps can be automated with Budibase (although, you are still required to buy a domain).

Choosing one of these hosting options will almost certainly provide you with everything you need. They have ample documentation and community support, and are generally reliable options.

Step 12 - Deploy your web app

You have sourced your idea, validated it, designed and developed your web app, and chosen your hosting provider. You're now at the last step.

The deployment step includes how your web application gets from your source control on your computer to your cloud hosting from step 11.

The following development tools provide continuous integration and will help you with deploying your web app to your cloud hosting:

1. GitLab
2. Bitbucket
3. Jenkins

Paper 3 of 5 — CSC 412 - 2025 handout

Level 400 • First Semester • Handout

CSC 412: NET – CENTRIC COMPUTING (3 UNITS)

COURSE OUTLINE

Distributed computing, mobile and wireless computing, network security; client/ server computing (using the web), building web applications.

DISTRIBUTED COMPUTING

Distributed computing is a method of using multiple computers to work together to solve a problem. It makes a network of computers appear as a single computer with large-scale resources.

Distributed computing refers to a system where processing and data storage is distributed across multiple devices or systems, rather than being handled by a single central device. In a distributed system, each device or system has its own processing capabilities and may also store and manage its own data. These devices or systems work together to perform tasks and share resources, with no single device serving as the central hub.

Examples of distributed computing

- Telecommunications networks: Telephone and cellular networks
- Network applications: The World Wide Web and peer-to-peer networks
- Real-time process control: Aircraft control systems
- Security cameras: Cameras send video data to a computer cluster for processing.
- Cloud computing: Services like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform
- Grid computing: A group of computers connected by networks that work together to solve a complex problem.

Components of distributed system.

There are several key components of a Distributed Computing System

- Devices or Systems: The devices or systems in a distributed system have their own processing capabilities and may also store and manage their own data.
- Network: The network connects the devices or systems in the distributed system, allowing them to communicate and exchange data.
- Resource Management: Distributed systems often have some type of resource management system in place to allocate and manage shared resources such as computing power, storage, and networking.

Characteristics of distributed computing system.

There are several characteristics that define a Distributed Computing System.

- Multiple Devices or Systems: Here, Processing and data storage is

distributed across multiple devices or systems.

- **Peer-to-Peer Architecture:** Devices or systems in a distributed system can act as both clients and servers, as they can both request and provide services to other devices or systems in the network.
- **Shared Resources:** Resources such as computing power, storage, and networking are shared among the devices or systems in the network.
- **Horizontal Scaling:** Scaling a distributed computing system typically involves adding more devices or systems to the network to increase processing and storage capacity. This can be done through hardware upgrades or by adding additional devices or systems to the network.

Advantages and Disadvantages of the distributed computing system.

Some Advantages of the Distributed Computing System are:

- **Scalability:** Distributed systems are generally more scalable than centralized systems, as they can easily add new devices or systems to the network to increase processing and storage capacity.
- **Reliability:** Distributed systems are often more dependable than centralized systems, as they can continue to operate even if one device or system fails.
- **Flexibility:** Distributed systems are generally more flexible than centralized systems, as they can be configured and reconfigured more easily to meet changing computing needs.

There are a few limitations to Distributed Computing System

- **Complexity:** Distributed systems can be more complex than centralized systems, as they involve multiple devices or systems that need to be coordinated and managed.
- **Security:** It can be more challenging to secure a distributed system, as security measures must be implemented on each device or system to ensure the security of the entire system.
- **Performance:** Distributed systems may not offer the same level of performance as centralized systems, as processing and data storage is distributed across multiple devices or systems.

Applications

Distributed Computing Systems have a number of applications, including:

- **Cloud Computing:** Cloud Computing systems are a type of distributed computing system that are used to deliver resources such as computing power, storage, and networking over the Internet.
- **Peer-to-Peer Networks:** Peer-to-Peer Networks are a type of distributed computing system that is used to share resources such as files and computing power among users.
- **Distributed Architectures:** Many modern computing systems, such as micro services architectures, use distributed architectures to distribute

processing and data storage across multiple devices or systems.

MOBILE & WIRELESS COMPUTING.

MOBILE COMPUTING

Mobile computing is the use of a computer while moving around, allowing users to access information and perform tasks without being tied to a single location. It involves the transmission of data, voice, and video using a wireless device.

Mobile computing refers to the set of IT technologies, products, services and operational strategies and procedures that enable end users to access computation, information and related resources and capabilities while mobile.

Mobile Computing is a technology that allows transmission of data, voice and video via a computer or any other wireless enabled device without having to be connected to a fixed physical link.

Components of mobile computing

They are:

Hardware: Includes computing silicon, sensors, and input/output devices.

These devices will have a receptor medium that is capable of sending and receiving signals. These devices are configured to operate in full- duplex, whereby they are capable of sending and receiving signals at the same time. They don't have to wait until one device has finished communicating for the other device to initiate communications.

Software: Includes programs that communicate with hardware, device drivers, applications, and software stacks. Mobile software is the actual program that runs on the mobile hardware. It deals with the characteristics and requirements of mobile applications. This is the engine of the mobile device. In other terms, it is the operating system of the appliance. It's the essential component that operates the mobile device.

Since portability is the main factor, this type of computing ensures that users are not tied or pinned to a single physical location, but are able to operate from anywhere. It incorporates all aspects of wireless communications.

Communication protocols: Includes Wi-Fi protocols and hypertext transfer protocol (HTTP). The mobile communication in this case, refers to the infrastructure put in place to ensure that seamless and reliable communication goes on. These would include devices such as protocols, services, bandwidth, and portals necessary to facilitate and support the stated services. The data format is also defined at this stage. This ensures that there is no collision with other existing systems which offer the same

service.

Mobile communication can be divided in the following four types:

1. Fixed and Wired
2. Fixed and Wireless
3. Mobile and Wired
4. Mobile and Wireless

Fixed and Wired: In Fixed and Wired configuration, the devices are fixed at a position, and they are connected through a physical link to communicate with other devices.

For Example, Desktop Computer.

Fixed and Wireless: In Fixed and Wireless configuration, the devices are fixed at a position, and they are connected through a wireless link to make communication with other devices.

For Example, Communication Towers, Wi-Fi router

Mobile and Wired: In Mobile and Wired configuration, some devices are wired, and some are mobile. They altogether make communication with other devices.

For Example, Laptops.

Mobile and Wireless: In Mobile and Wireless configuration, the devices can communicate with each other irrespective of their position. They can also connect to any network without the use of any wired device.

For Example, Wi-Fi Dongle.

ADVANTAGES AND DISADVANTAGES OF MOBILE COMPUTING.

Advantages of computing.

- **Portability and Flexibility:** One of the biggest advantages of mobile computers is their portability. Whether you're a student, a professional, or someone who loves to travel, having a device you can carry with ease is a game-changer.
- **Increased Productivity:** Mobile computers help users stay productive, even when they're not in the office or at home. With cloud computing and remote access, you can manage tasks, attend meetings, and share files seamlessly.
- **Enhanced Communication:** Mobile computers make it easier to communicate in real-time through emails, instant messaging, and video conferencing.
- **Versatile:** You can utilize your gadget while strolling or dozing in your room. In the event that you are going in a vehicle, at that point you can in any case make live video and offer with your companions via web-based media. You can likewise live stream at any show utilizing your cell phone.

- **Better Time the Executives:** By utilizing a cell phone or tablet PC you can introduce any time following an application by which that application can more readily deal with your time. There are numerous applications accessible that can assist you with getting sorted out yourself. Some applications block some social applications that burn through your time. You can put together your time and spotlight on things that issue to you.
- **Grow Your Business:** You can work together by utilizing a cell phone moreover. You can run Facebook advertisements, oversee Facebook business pages, or some other social movement you can do on your portable. On the off chance that you have cameras in your organization, at that point you can screen the presentation of your representative's privilege from your portable.
- **Checking Climate and Securities Exchange:** On the off chance that you have the arrangement to go on the outing, at that point checking the climate is significant. By utilizing a portable PC you can check the climate without any problem. In the event that you are a stockbroker, at that point checking stock costs is important. You can likewise exchange your stocks while utilizing your cell phone.
- **Simple to Convey:** You can convey your portable PC without any problem. You can keep it in your pocket or pack moreover. It is additionally lightweight than office PC. You can accomplish your office work like checking the bookkeeping pages, messages, or any talks from your customers.

Disadvantages of Mobile Computer

- **Availability Issues:** You need to interface with the web for utilizing any applications and it needs Wi-Fi or a versatile information network. On the off chance that you are in an area where versatile signs are feeble, at that point it makes an issue. Likewise, an awful climate can detach you from the web. Your portable will likewise be charged/charged in the event that you need to utilize the web on your gadget.
- **Battery Issue:** All cell phones utilize a battery for charging your gadget. In the event that you have no spot to charge portable battery for example you are away from home then it makes an issue. You can't utilize your cell phone if your battery isn't charged.
- **Security:** In the event that you are associated with the web, at that point, you are dangerous as your information can be taken. Some applications can follow your area and search terms you use on various sites. On the off chance that you utilize your credit/check cards online then your cash is likewise in danger. Programmers can hack your versatile/tablet/PC and get your secret information from your gadget. So it is smarter to not store any of your bank

subtleties, credit/charge card subtleties on your portable.

- Insurance: As cell phones are expensive as they have opportunities to be taken. They additionally have opportunities to tumbled from your hand because of little size. Some cell phone and tablets are exorbitant to such an extent that you feel perilous to utilize them out in the open spots.
- Limited Battery Life: While portability is a major benefit, it comes at the cost of limited battery life. Long usage periods without access to charging can be problematic, especially in professional settings.
- Security Vulnerabilities: Mobile devices are often more susceptible to security risks such as malware, hacking, or data theft, especially when connected to public networks. Example: A hacker could intercept sensitive information when you use an unsecured Wi-Fi network.
- Performance Limitations: Compared to desktop computers, mobile computers may not have the same processing power or storage capacity, which can limit their use for resource-heavy applications.

APPLICATIONS OF MOBILE COMPUTING

- Internet access: Users can access the internet on their mobile devices using Wi-Fi or other wireless networks.
- Global Positioning System (GPS): Users can use GPS on their mobile devices to navigate.
- Emergency services: Mobile devices can be used to contact emergency services.
- Entertainment services: Mobile devices can be used to access entertainment services.
- Educational services: Mobile devices can be used to access educational services.
- Remote work: Users can work remotely using mobile devices.
- Resource access: Users can access resources from different locations using mobile devices.
- Internet of Things (IoT): Mobile devices can be used to connect to and control IoT devices, such as smart home appliances, wearables, and transportation systems.
- Data access: Mobile devices can be used to access data stored in the cloud.
- Communication: Mobile devices can be used to communicate with others.
- Quality control: Mobile devices can be used to improve quality control in manufacturing.
- Data-driven decision making: Mobile devices can be used to make data-driven decisions.

WIRELESS COMPUTING.

Wireless computing is the use of wireless devices and networks to access the internet and intranet without cables. Wireless computing allows for portable, immediate, and accurate interactions.

Wireless computing uses radio frequency (RF) connections to send data between devices.

Devices can communicate directly with each other or connect to a network through a wireless access point (AP).

Wireless networks are used in homes, businesses, and telecommunications networks.

some examples of wireless computing

- Wi-Fi (wireless fidelity)

A popular wireless networking technology that uses radio frequency (RF) connections to create networks in homes, businesses, and telecommunications networks

- Bluetooth

A wireless technology that allows Bluetooth-enabled devices to communicate with each other over short distances.

- Cellular data services

A type of wireless computing that allows users to make phone calls and access data over a cellular network.

- Mobile satellite communications

A type of wireless computing that allows users to communicate with each other using satellites.

Advantages of wireless computing.

1. Freedom from wires: Can be configured without the use of any physical connection.
2. Easy to setup: Wireless network is easy to expand and setup
3. Better or global coverage: It provides global reach by providing networking in places such as rural areas, battlefields, etc... where wiring is not feasible.
4. Flexibility: Wireless network is more flexible and adaptable compared to a wired network.
5. Cost-effectiveness: Since it is easy to install and doesn't require cables, the wireless network is relatively cheaper.
6. Mobile and portable: Wireless network is easy to carry and re-install in another place.
7. Mobility -: It has good mobility of usage.

8. Data transmission is fast: In wireless network data transmission is fast.
9. Low maintenance: In any wireless communication low maintenance cost.
10. Network planning: Network planning in wireless communication is very easy as compare to wired network due to wireless software configuration of frequency, power and other parameter.
11. Expandable: Wireless networks can serve a large number of client without resorting the new hardware.

Disadvantages of wireless computing.

1. As communication is done through open space, it is less secure.
2. Unreliability
3. More open to interference.
4. Increased chance of jamming.
5. Transmission speed is comparably less.
6. it has a limited amount of bandwidth for communication and breaches of network security.
7. Wireless networks can be easily hacked.
8. Wireless networks require a careful radio frequency when they are installed.
9. Wireless networks are usually inexpensive, but the cost of installation is very high, setting up a wireless network is very costly.
10. Difficult to set up little experience people.

Applications of Wireless Communication:

1. Satellite system
2. Television remote control
3. Wi-Fi
4. Paging system
5. Wi-Max
6. Security systems
7. Cellphones
8. Computer interface devices
9. Bluetooth
10. GPS
11. GSM
12. Accessing the internet.
13. For locating and tracing someone.
14. Wireless power transfer.

NETWORK SECURITY

Any action intended to safeguard the integrity and usefulness of your data and network is known as network security. In other words, Network security is defined as the activity created to protect the integrity of your network and data.

Network security is the practice of protecting a computer network from unauthorized access, misuse, or attacks. It involves using tools, technologies, and policies to ensure that data traveling over the network is safe and secure, keeping sensitive information away from hackers and other threats.

How Does Network Security Work?

Network security uses several layers of protection, both at the edge of the network and within it. Each layer has rules and controls that determine who can access network resources. People who are allowed access can use the network safely, but those who try to harm it with attacks or other threats are stopped from doing so.

The basic principle of network security is protecting huge stored data and networks in layers that ensure the bedding of rules and regulations that have to be acknowledged before performing any activity on the data.

These levels are:

Physical Network Security: This is the most basic level that includes protecting the data and network through unauthorized personnel from acquiring control over the confidentiality of the network. The same can be achieved by using devices like biometric systems.

Technical Network Security: It primarily focuses on protecting the data stored in the network or data involved in transitions through the network. This type serves two purposes. One is protected from unauthorized users, and the other is protected from malicious activities.

Administrative Network Security: This level of network security protects user behavior like how the permission has been granted and how the authorization process takes place. This also ensures the level of sophistication the network might need for protecting it through all the attacks. This level also suggests necessary amendments that have to be done to the infrastructure.

Types of Network Security

There are several types of network security through which we can make our network more secure, your network and data are shielded from breaches, invasions, and other dangers by network security. Here below are some important types of network security:

Email Security

Email Security is defined as the process designed to protect the Email Account and its contents safe from unauthorized access. For Example, you generally see,

fraud emails are automatically sent to the Spam folder. because most email service providers have built-in features to protect the content.

The most common danger vector for a security compromise is email gateways. Hackers create intricate phishing campaigns using recipients' personal information and social engineering techniques to trick them and direct them to malicious websites. To stop critical data from being lost, an email security program restricts outgoing messages and stops incoming threats.

Network Segmentation

Network traffic is divided into several categories by software-defined segmentation, which also facilitates the enforcement of security regulations. Ideally, endpoint identity—rather than just IP addresses—is the basis for the classifications. To ensure that the appropriate amount of access is granted to the appropriate individuals and that suspicious devices are controlled and remediated, access permissions can be assigned based on role, location, and other factors.

Access Control

Your network should not be accessible to every user. You need to identify every user and every device in order to keep out any attackers. You can then put your security policies into effect. Noncompliant endpoint devices might either have their access restricted or blocked. Network access control (NAC) is this process.

Sandboxing

Sandboxing is a cybersecurity technique in which files are opened or code is performed on a host computer that simulates end-user operating environments in a secure, isolated environment. To keep threats off the network, sandboxing watches the code or files as they are opened and searches for harmful activity.

Cloud Network Security

This is very vulnerable to the malpractices that few unauthorized dealers might pertain to. This data must be protected and it should be ensured that this protection is not jeopardized by anything. Many businesses embrace SaaS applications for providing some of their employees the allowance of accessing the data stored in the cloud. This type of security ensures creating gaps in the visibility of the data.

Workloads and applications are no longer solely housed in a nearby data center on-site. More adaptability and creativity are needed to protect the modern data center as application workloads move to the cloud.

Web Security

An online security solution will restrict access to harmful websites, stop web-based risks, and manage staff internet usage. Your web gateway will be safeguarded both locally and in the cloud. "Web security" also include the

precautions you take to safeguard your personal website.

Intrusion Prevention System(IPS)

An intrusion Prevention System is also known as Intrusion Detection and Prevention System. It is a network security application that monitors network or system activities for malicious activity. The major functions of intrusion prevention systems are to identify malicious activity, collect information about this activity, report it, and attempt to block or stop it.

Antivirus and Anti-Malware Software

This type of network security ensures that any malicious software does not enter the network and jeopardize the security of the data. Malicious software like Viruses, Trojans, and Worms is handled by the same. This ensures that not only the entry of the malware is protected but also that the system is well-equipped to fight once it has entered.

Firewalls Security

A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules accepts, rejects, or drops that specific traffic. Before Firewalls, network security was performed by Access Control Lists (ACLs) residing on route

Application Security

Application security denotes the security precautionary measures utilized at the application level to prevent the stealing or capturing of data or code inside the application. It also includes the security measurements made during the advancement and design of applications, as well as techniques and methods for protecting the applications whenever.

Wireless Security

Wireless networks are less secure than wired ones. If not properly secured, setting up a wireless LAN can be like having Ethernet ports available everywhere, even in places like parking lots. To prevent attacks and keep your wireless network safe, you need dedicated products designed to protect it from exploits and unauthorized access.

Web Security

A web security solution manages how your staff uses the internet, blocks threats from websites, and stops access to harmful sites. It safeguards your web gateway either onsite or in the cloud. Additionally, “web security” involves measures taken to protect your own website from potential attacks and vulnerabilities.

Mobile Device Security

Cybercriminals are focusing more on mobile devices and apps. In the next three years, about 90 percent of IT organizations might allow corporate applications on personal mobile devices. It's crucial to control which devices can connect to your

network and set up their connections securely to protect network traffic from unauthorized access.

Industrial Network Security

As industries digitize their operations, the closer integration of IT, cloud services, and industrial networks exposes Industrial Control Systems (ICS) to cyber threats. To safeguard against these risks, it's crucial to have complete visibility into your Operational Technology (OT) security status. This involves segmenting the industrial network and providing detailed information about OT devices and their behaviors to IT security tools. This approach helps in effectively monitoring and protecting critical industrial systems from potential cyber-attacks.

VPN Security

A virtual private network (VPN) encrypts the connection between a device and a network, usually over the internet. A remote-access VPN commonly uses IPsec or Secure Sockets Layer (SSL) to verify and secure the communication between the device and the network. This encryption ensures that data transmitted between the device and the network remains private and secure from unauthorized access.

Benefits of Network Security

Network Security has several benefits, some of which are mentioned below:

Network Security helps in protecting clients' information and data which ensures reliable access and helps in protecting the data from cyber threats.

Network Security protects the organization from heavy losses that may have occurred from data loss or any security incident.

It overall protects the reputation of the organization as it protects the data and confidential items.

Advantages of Network Security

Protection from Unauthorized Access: Network security measures such as firewalls and authentication systems prevent unauthorized users from accessing sensitive information or disrupting network operations.

Data Confidentiality: Encryption technologies ensure that data transmitted over the network remains confidential and cannot be intercepted by unauthorized parties.

Prevention of Malware and Viruses: Network security solutions like antivirus software and intrusion detection systems (IDS) detect and block malware, viruses, and other malicious threats before they can infect systems.

Secure Remote Access: Virtual private networks (VPNs) and other secure remote access methods enable employees to work remotely without compromising the security of the organization's network and data.

Disadvantages of Network Security

Complexity and Management Overhead: Implementing and managing network security measures such as firewalls, encryption, and intrusion detection systems (IDS) can be complex and require specialized knowledge and resources.

Cost: Effective network security often requires investment in hardware, software, and skilled personnel, which can be expensive for organizations, especially smaller ones.

Privacy Concerns: Some network security measures, such as deep packet inspection and monitoring, may raise privacy concerns among users and stakeholders, requiring careful balancing of security needs with individual privacy rights.

CLIENT AND WEB COMPUTING USING WEB

"Client/server computing using the web" refers to a system where a user's device (the "client," like a web browser) sends requests for information or services to a remote computer (the "server") over the internet, which then processes the request and sends back the requested data, allowing users to access web pages, applications, and other online resources; essentially, the standard way most people interact with websites and online services.

Key points about client/server computing on the web:

Client:

The user's device, typically a computer or mobile phone, with a web browser that initiates requests to the server.

Server:

A computer on the internet that stores web pages and data, and is responsible for processing client requests and sending back responses.

Communication Protocol:

HTTP (Hypertext Transfer Protocol) is the standard protocol used for communication between the client and server.

How it works:

1. User Action:

The user types a URL into their web browser.

2. DNS Lookup:

The browser sends a request to a DNS server to resolve the URL into an IP address.

3. Request Sent:

The browser sends an HTTP request to the web server at the identified IP address.

4. Server Processing:

The web server receives the request, retrieves the requested data from its storage, and processes it.

5. Response Sent:

The server sends back the requested data (e.g., HTML, CSS, images) to the browser in the form of an HTTP response.

6. Display on Browser:

The browser interprets the received data and displays the webpage to the user.

Examples of client/server web applications:

- Accessing a website:

When you visit a website like Google or Facebook, your browser (client) sends a request to the website's server, which then sends back the webpage content.

- Online banking:

When you log into your online banking account, your browser sends login credentials to the bank's server, which verifies the information and provides access to your account.

- Email:

Sending and receiving emails involves your email client (like Gmail or Outlook) communicating with an email server to send and retrieve messages.

In client server computing, the clients request a resource and the server provides that resource. A server may serve multiple clients at the same time while a client is in contact with only one server. Both the client and server usually communicate via a computer network but sometimes they may reside in the same system.

Characteristics of Client Server Computing

The salient points for client server computing are as follows:

- The client server computing works with a system of request and response.

The client sends a request to the server and the server responds with the desired information.

- The client and server should follow a common communication protocol so they can easily interact with each other. All the communication protocols are available at the application layer.

- A server can only accommodate a limited number of client requests at a time. So it uses a system based to priority to respond to the requests.

- An example of a client server computing system is a web server. It returns the web pages to the clients that requested them.

Advantages of Client Server Computing

The different advantages of client server computing are –

- All the required data is concentrated in a single place i.e. the server. So it is

easy to protect the data and provide authorisation and authentication.

- The server need not be located physically close to the clients. Yet the data can be accessed efficiently.
- It is easy to replace, upgrade or relocate the nodes in the client server model because all the nodes are independent and request data only from the server.
- All the nodes i.e. clients and server may not be built on similar platforms yet they can easily facilitate the transfer of data.

Disadvantages of Client Server Computing

The different disadvantages of client server computing are –

- If all the clients simultaneously request data from the server, it may get overloaded. This may lead to congestion in the network.
- If the server fails for any reason, then none of the requests of the clients can be fulfilled. This leads of failure of the client server network.
- The cost of setting and maintaining a client server model are quite high.

BUILDING WEB APPLICATIONS

A web application or web app is a software application that can be accessible via a web browser or a web-enabled device, such as a smartphone or tablet. It operates on a web server and anyone with an internet connection can use it.

Building a web application involves defining the app's purpose, researching the market, and designing the app's user interface and database. It also requires planning the workflow, creating a prototype, and testing the app before launching it.

Steps for building a web application

1. Define the app's purpose: Consider what the app's goals are, such as improving customer engagement or streamlining internal operations.
2. Research the market: Identify your target users and competitors.
3. Plan the workflow: Consider how you'll use your resources and budget to meet the app's requirements.
4. Create a prototype: Use wireframes and mockups to visualize the app's structure and layout.
5. Design the database: Choose a database system and design an efficient schema.
6. Build the frontend and backend: Use graphics, templates, and style guidelines to design the app's interactive components.
7. Test and troubleshoot: Thoroughly test the app before launching it.
8. Deploy the app: Choose a hosting provider and transfer the app to the cloud.

Top 10 Tools for Modern Web Building/Development for beginners

1. Visual Studio Code (VS Code)

VS Code probably is the most used code editor among web developers. In fact, it was made by Microsoft: lightweight, fast, and highly extensible with a vast extension ecosystem. VS Code supports a wide variety of programming languages natively-from JavaScript and TypeScript to Python, HTML, and CSS. It can be used both for the development of the frontend and the backend.

Key Features:

- **Integrated Git Support:** Managing Git and versioning of source code is pretty much straightforward and easy, as all the Git commands come integrated right within VS Code.
- **IntelliSense:** Provides intelligent code completion, parameter information, and real time error checking.
- **Extensions:** An ecosystem rich in extensions doing everything from debugging through to linting and testing.
- **Multilanguage Support:** It supports many of a wide variety of programming languages to several frameworks.

2. Webpack

Webpack is a very powerful module bundler for JavaScript applications. This tool pre-processes and bundles different assets of an application, such as JavaScript, CSS, HTML, images, into various optimized production-ready files. Webpack helps developers maintain their dependency graph and efficiently load their resources while improving developer workflows and application performance.

Key Features:

- **Code Splitting:** It bifurcates your code into tiny pieces, reducing the load time.
- **Hot Module Replacement (HMR):** Updates parts of your app in real-time without refreshing the whole page, hence fastening up the development process.
- **Plugins and Loaders:** In this article, the support for a variety of plugins and loaders for handling CSS, images, fonts, and other assets is supported.
- **Tree Shaking:** It automatically removes unused code to reduce the final bundle size.

3. Node.js

Node.js is a JavaScript runtime; built on Chrome's V8 JavaScript Engine, it allows developers to run JavaScript on the server-side. It sees huge usage in the development of fast, scalable, and real-time applications, be it chat apps, APIs, SPAs, and so on.

Key features:

- **Asynchronous Event-Driven Architecture:** It can manage several requests without blocking the thread and hence finds its application in developing high-performance-driven applications.
- **NPM:** Node Package Manager actually provides you with the largest package ecosystem, allowing developers to quickly implement third-party libraries and frameworks.
- **Cross-platform:** Works on Windows, macOS, and Linux, enabling developers to create applications which can run on each of these systems.
- **Scalability:** The framework can easily be applied to both smaller and larger-scale web application developments.

4. React

React is an open-source, front-end JavaScript library developed by Facebook for designing user interfaces, and more importantly and most commonly single-page applications. It enables one to create reusable UI components that maintain their own state, providing code that is both predictable and maintainable. The component-based architecture in React has made it one of the most favored front-end tools for modern web development.

Key Features:

- **Virtual DOM:** The keyword virtual DOM in React provides the ability to the Runtime, where only the part changed by the user is updated and the rest of the UI remains as it was for better performance.
- **Component-Based Architecture:** It encourages the construction of modular and reusable building blocks, which foster cleaner and better-maintained code.
- **JSX Syntax:** It has combined syntactical features from JavaScript and HTML in one, making very effective the creation of dynamic UI.
- **Rich Ecosystem:** Strong ecosystem of tools, libraries, and extension points to get your projects going, including React Router, Redux, and Material-UI.

5. GitHub

GitHub is a free-of-cost, web-based platform for version control and collaboration built upon the Git tool. It is very popularly utilized for maintaining code repositories, tracking ongoing changes in the same, and collaborating with fellow buggy developers on projects. More than that, it has large numbers of open-source projects hosted, and contributes to community-driven codebases.

Key Features:

- **Version Control:** Version control keeps a record of all the modifications that happened to your code over time and allows you to go back at any time to the previous versions.

- Collaboration: Will allow collaboration between teams with pull requests, code review, and branch management.
- GitHub Actions: Automate your CI/CD workflows directly in your repository.
- Open-Source Community: Lots of opensource projects are hosted here for developers to contribute or get inspiration from.

6. Postman

Postman is one of the API development and testing utilities helping to simplify this work with APIs. It allows creating, testing, and documenting APIs much more easily. Thus, for a backend and full-stack developer who often has to interact with APIs, Postman proves to be a necessary utility tool.

Key Features:

- API testing: Perform GET, POST, PUT, DELETE, and any other type of HTTP request with just a couple of clicks.
- Automated Testing: It does support automated testing, thus the integration of APIs with continuous integration/continuous deployment, that is, CI/CD, to make sure it's all working as it should.
- API Documentation: It enables developers to generate maintainable API documentation in a readable format.
- Collaboration: allows sharing API collections for teams to collaborate in API development.

7. Sass

Sass stands for Syntactically Awesome Stylesheets. It is a CSS preprocessor augmenting CSS by adding features such as variables, nested rules, mixins, and functions, among others. It helps the developer in writing more maintainable and scalable CSS code, especially for larger projects involving complex style rules.

Key Features:

- Variables: For the use of variables to store values - colors, fonts, among other reusable properties.
- Nesting: Allows CSS selector nesting, which makes stylesheets more readable and, thus, easier to maintain.
- Mixins: Allows reutilization of CSS declarations, so in return reducing redundancy and making the code more maintainable.
- Partial and Imports: Partial and imports help with large stylesheets by breaking them down into smaller, maintainable files.

8. ESLint

ESLint is a tool for linting JavaScript code; it highlights problems and fixes code. The substantial idea behind it is to maintain conventions about coding and best

practices by keeping consistency among teams and projects. ESLint can be configured high, supports the addition of custom rules, and therefore has found its place in the MUST-HAVE toolkit for a modern web developer.

Key Features:

- **Code Quality Enforcement:** Points out syntax errors, coding style violations, and potential bugs in JavaScript.
- **Configurable:** It enables the developers to specify rules for linting and thus reach their desired coding style, along with meeting project requirements.
- **Editor Integration:** Integrates with the likes of VS Code and Sublime Text to offer real-time linting feedback.
- **Extensibility:** It runs plugins and extensions by default, which enforce rules for specific frameworks like React or Node.js.

9. Figma

Figma is an online design and prototyping tool that allows designers and developers to collaborate in real time. It is one of the most used tools for designing web interfaces, creating interactive prototypes, and sharing design specifications with development teams. Because of its collaborative features, Figma is very apt for any team working on any web development project.

Key features:

- **Real-time collaboration:** It means allowing more than one user to work on the same design project simultaneously.
- **Design prototyping:** It allows designers to create interactive prototypes that simulate the user experience.
- **Developer Handoff:** It helps developers with design specs, assets and code snippets to make the handoff process smooth.
- **Cross-Platform:** Since it works on any operating system with an installed web browser, it is highly accessible to teams.

10. Jest

Jest is a JavaScript test runner from Facebook that helps make sure the code works as one would expect. Jest has broad usage in testing React apps, but it is far from limited to that use. It can be used to test JavaScript and TypeScript in other environments. Jest has an easy API to write unit tests, integration tests and snapshots.

Key Features:

- **Fast and Reliable:** Jest runs tests in parallel, hence fast execution of tests and sure results.
- **Snapshot Testing:** This allows you to capture the rendered output of your components and compare them over time.

- Mocking: Supports mock for the isolation of code when testing.
- Watch Mode: Automatically re-runs tests when files are modified, making it easier to catch bugs during development.

Top 10 Web Development Tools for Advanced Developers

1. Postman
2. Docker
3. Kubernetes
4. Sketch
5. NGINX
6. Flutter
7. ReactJS
8. Angular
9. Vue.js
10. Laravel

Paper 4 of 5 — CSC 412 handout

Level 400 • First Semester • Handout

Lecture notes on

Net-centric Computing

Csc 412

FIRST SEMESTER (2023/2024 SESSION)

DEPARTMENT OF COMPUTER SCIENCE,

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COURSE OUTLINE:-

- Introduction to Net-Centric Computing
- Distributed Computing
- Mobile and Wireless Computing
- Network Security
- Client/Server Computing (using the web)
- Building Web Applications

COURSE summary:-

The goal of this course is to familiarize the student with the fundamentals of computer networks, network types and categories, distributed computing, dynamic devices and mode of transmission, security of connected and communicating systems using ciphers, Client and Servers communication, and building of web applications. Upon completing this course, you would have been equipped with the knowledge of Net-centric computing fundamentals, what network is all about and wireless technologies, cloud computing and service models, and much more.

Lecture one

Introduction to

Net-Centric Computing

For several years, major changes are being brought to the world by universal networking capabilities, such as the Internet. Today's technology solutions represent the convergence of computing power, networking capability and the information, data or knowledge that forms the content of these solutions. At the centre these solutions net-centric computing lies.

Net-centric computing refers to an emerging technology architecture for modern distributed computing systems and applications, and an evolutionary stage of client/server computing. Its architecture is built on open standards that supports different people to collaborate and to reach different information sources in different ways. The evolutionary nature of net-centric computing links, technological capabilities and strategic opportunities, thus helping people in facing today's new problems, and providing the flexibility to meet tomorrow's challenges.

Definition: Net-Centric Computing (NCC) is a distributed environment where applications and data are downloaded from servers and exchanged with peers across a network. Net-centric

Computing focuses on large-scale distributed computing systems and applications that communicate through open, wide-area networks like the Internet. General examples of large-scale network-centric systems are the World-Wide Web and Computational Grids.

Net-Centric Computing itself is the future of computing that brings together the lives of people. The earliest computer systems functioned in isolation from one another; but today it is almost impossible to find a computer system that is not networked in some way to other systems. This includes not only desktop and laptop computers, but also devices such as cell phones, entertainment devices and this will face massive changes in the future with advancing technologies like IoT (Internet of Things) and AI (Artificial Intelligence).

OBJECTIVES OF NET-CENTRIC COMPUTING

- Net-centric computing refers to a computing paradigm that emphasizes the use of networked systems and services to enable communication, collaboration, and information sharing.
- It involves the integration of various technologies, such as computer networks, internet protocols, distributed computing, and web-based applications, to create a seamless and interconnected computing environment.
- It focuses on leveraging the power of the network to enable efficient and effective sharing of resources, data, and services. This approach allows users to access and utilize computing resources and services from anywhere, at any time, using any device with network connectivity.
- It enables the development and deployment of distributed applications and services that can be accessed and used by multiple users simultaneously. It promotes interoperability and integration among different systems and platforms, allowing for seamless communication and collaboration across organizational boundaries.

KEY CONCEPTS IN NET-CENTRIC COMPUTING

1. Service-oriented architecture (SOA): This architectural approach emphasizes the creation of modular and loosely coupled services that can be easily combined and reused to build complex applications.
2. Cloud computing: This refers to the delivery of computing resources, such as storage, processing power, and software applications, over the internet on a pay-as-you-go basis. It enables scalability, flexibility, and cost-efficiency in deploying and managing applications.
3. Web services: These are software components that expose their functionality over the internet using standard protocols, such as HTTP and XML. Web services enable interoperability and integration among different systems and platforms.
4. Big data: Net-centric computing deals with the processing and analysis of large volumes of data generated from various sources, such as social media, sensors, and transactional systems. It involves techniques for data storage, retrieval, analysis, and visualization.

WHAT ARE THE BENEFITS OF NET-CENTRIC COMPUTING?

- NCC provides the foundation for the development of innovative applications and services

that leverage the power of the network to enhance productivity, efficiency, and effectiveness in various domains.

□ NCC can be considered an evolution of Software as a Service (SaaS). It allows you to manage your data, applications, and infrastructure from one interface (the cloud) without worrying about hardware or server management issues.

□ NCC combines network-based computing with its integration of various types of information technology resources (servers, storage devices, servers, computers) into centralized repositories that are served using standard Web-based protocols such as HTTP or HTTPS via a global computer communications network like the internet.

□ NCC allows organizations to focus on their core business needs without limiting themselves by software or hardware limitations imposed on their infrastructure. In other words, when an organization adopts net-centric principles, they are able to completely virtualize its IT footprint while still being able to take advantage of modern networking technologies like Local Area Networks (LANs) and Wide Area Networks (WANs).

□ NCC is a combination of IaaS, PaaS, and SaaS. What this means is that instead of buying hardware and software for your own data centre, you buy it from the cloud provider. This gives you the ability to move your data to the cloud and access it from anywhere.

□ NCC services allows you to centralize your applications with a single interface. It provides fully managed services according to user's specific requirements, which are invoked in real-time as needed rather than being provided on-demand or already provisioned for use. This concept enables multiple distributed clients to access a single entity's application in real-time.

WHAT ARE THE CAREERS RELATED TO NET-CENTRIC COMPUTING?

NCC covers a range of specialties within computing, including: computer communication, network concepts and protocols, web standards and technologies, network security, wireless and mobile computing, and distributed systems development (including web-based systems).

A wide range of careers related to net-centric computing include but are not limited to;

- Network Administrator □ Wireless Networking Engineer
- Network Engineer □ Information Security Analyst
- Security Architect □ Web Applications Developer
- Systems Analyst □ Cloud Networking Architect
- Systems designer □ Java and/or .NET developer
- Systems developer □ Mobile applications developer

Lecture two

cloud Computing

Cloud computing is a general term for anything that involves delivering hosted services over the internet in order to provide easy, scalable access to computing resources and IT services. Gone are the days when businesses were compelled to store or access programs, apps directly on their

hard drive. Today we have cloud technology.

In simple terms, cloud computing allows you to rent instead of buy your IT. Rather than investing heavily in databases, software, and hardware, companies opt to access their compute power via the internet, or the cloud, and pay for it as they use it. These cloud services now include, but are not limited to, servers, storage, databases, networking, software, analytics, and business intelligence.

Definition: Cloud computing refers to the use of hosted services, such as data storage, servers, databases, networking, and software over the internet. The data is stored on physical servers, which are maintained by a cloud service provider. Computer system resources, especially data storage and computing power, are available on-demand, without direct management by the user in cloud computing.

CLOUD COMPUTING ARCHITECTURE

□ Instead of storing files on a storage device or hard drive, a user can save them on cloud, making it possible to access the files from anywhere, as long as they have access to the web.

□ The services hosted on cloud can be broadly divided into infrastructure-as-a-service (IaaS), platform-as-a-service (PaaS), and software-as-a-service (SaaS). Based on the deployment model, cloud can also be classified as public, private, and hybrid cloud.

□ Further, cloud can be divided into two different layers, namely, front-end and back-end.

o The layer with which users interact is called the front-end layer. This layer enables a user to access the data that has been stored in cloud through cloud computing software.

o The layer made up of software and hardware components, i.e., the computers, servers, central servers, and databases, is the back-end layer. This layer is the primary component of cloud and is entirely responsible for storing information securely.

□ To ensure seamless connectivity between devices linked via cloud computing, the central servers use a software called middleware that acts as a bridge between the database and applications.

TYPES OF CLOUD COMPUTING

Cloud computing can either be classified based on the deployment model or the type of service. Based on the specific deployment model, we can classify cloud as public, private, and hybrid cloud. At the same time, it can be classified as infrastructure-as-a-service (IaaS), platform-as-a-service (PaaS), and software-as-a-service (SaaS) based on the service the cloud model offers.

BASED ON DEPLOYMENT

Private Cloud (also termed internal, enterprise, or corporate cloud) –

□ A private cloud is used exclusively by a single organization. It could be hosted at the organization's location or at the cloud provider's data centre.

□ Computing services are offered over a private IT network for dedicated use, and services

are usually managed via internal resources that is not accessible to anyone outside the organization.

- Private clouds provide a higher level of security through company firewalls and internal hosting to ensure that an organization's sensitive data is not accessible to third-party providers.

- The drawback of private cloud, however, is that the organization becomes responsible for all the management and maintenance of the data centers, which can prove to be quite resource-intensive.

- Some major private cloud vendors include Amazon Virtual Private Cloud (VPC), Hewlett Packard Enterprise (HPE), VMware, IBM Cloud, CISCO Quickstart Private Cloud, Dell EMC, Oracle Private Cloud Appliance X8, etc.

Public Cloud –

- In a public cloud, the entire computing infrastructure is located on the premises of the cloud provider, and the provider delivers services to the customer over the internet. Customers do not have to maintain their own IT and can quickly add more users or computing power as needed.

- Unlike private cloud, the services on public cloud are available to anyone who wants to use or purchase them. These services could be free or sold on-demand, where users only have to pay per usage for the CPU cycles, storage, or bandwidth they consume.

- Public clouds can help businesses save on purchasing, managing, and maintaining on-premises infrastructure since the cloud service provider is responsible for managing the system.

- They also offer scalable RAM and flexible bandwidth, making it easier for businesses to scale their storage needs.

- Some major public cloud providers include Google Workspace, Amazon Web Services (AWS), Microsoft offerings like Microsoft 365 and Azure, as well as streaming services.

Hybrid Cloud –

- As the name suggests, a hybrid cloud is a combination of both public and private clouds features. The “best of both worlds” cloud model allows a shift of workloads between private and public clouds as the computing and cost requirements change.

- Generally, hybrid cloud customers host their business-critical applications on their own servers for more security and control, and store their secondary applications at the cloud provider's location.

- When the demand for computing and processing fluctuates, hybrid cloud allows businesses to scale their on-premises infrastructure up to the public cloud to handle the overflow while ensuring that no third-party data centers have access to their data.

- In a hybrid cloud model, companies only pay for the resources they use temporarily instead of purchasing and maintaining resources that may not be used for an extended period. In short, a hybrid cloud offers the benefits of a public cloud without its security risks.

□ Some examples of hybrid cloud platforms include AWS Outposts, Azure Stack, Azure Arc, Azure VMware Solution, Google Anthos, Nutanix Cloud Infrastructure, Nutanix Cloud Clusters, VMware Cloud Foundation and VMware Cloud on AWS.

BASED ON SERVICE MODEL

Infrastructure as a Service (IaaS) -

□ Infrastructure as a service or IaaS is a type of cloud computing in which a service provider is responsible for providing servers, storage, and networking over a virtual interface. IaaS enables customers to access infrastructure services on an on-demand basis via the internet.

□ The key advantage is that the cloud provider hosts the infrastructure components that provide compute, storage, and network capacity so that subscribers can run their workloads in the cloud.

□ The cloud subscriber is usually responsible for installing, configuring, securing, and maintaining any software on the cloud native solutions, such as database, middleware, and application software.

Platform as a Service (PaaS) -

□ Platform as a service or PaaS is a type of cloud computing that provides a development and deployment environment in the cloud, which allows users to develop and run applications without the complexity of building or maintaining the infrastructure.

□ PaaS provides users the advantage of accessing the developer tools they need to build and manage mobile and web applications without investing in or maintaining the underlying infrastructure. The provider hosts the infrastructure and middleware components, and the customer accesses those services via a web browser. This allows organizations to focus on the deployment and management of their applications by liberating them from the responsibility of software maintenance, planning, and resource procurement.

□ To aid productivity, PaaS solutions need to have ready-to-use programming components that allow developers to build new capabilities into their applications, including innovative technologies such as artificial intelligence (AI), chatbots, blockchain, and the Internet of Things (IoT).

□ The right PaaS offering also should include solutions for analysts, end users, and professional IT administrators, including big data analytics, content management, database management, systems management, and security.

Software as a Service (SaaS) -

□ SaaS is a software delivery model in which the cloud provider hosts the customer's applications at the cloud provider's location. The customer accesses those applications over the internet. Rather than paying for and maintaining their own computing infrastructure, SaaS customers take advantage of subscription to the service on a pay-as-you-go basis.

□ In the SaaS model, the cloud provider manages all the hardware, middleware, application software, and security. Also referred to as 'hosted software' or 'on-demand software', SaaS makes it easy for enterprises to streamline their maintenance and support.

□ Many businesses find SaaS to be the ideal solution because it enables them to get up and running quickly with the most innovative technology available. Automatic updates reduce the burden on in-house resources. Customers can scale services to support fluctuating workloads, adding more services or features they grow.

□ A modern cloud suite provides complete software for every business need, including customer experience, customer relationship management, customer service, enterprise resource planning, procurement, financial management, human capital management, talent management, payroll, supply chain management, enterprise planning, and more.

□ Popular examples of SaaS applications include Zoom, Dropbox, Google Suite, Slack, Netflix, etc.

KEY BENEFITS

The most important reason why cloud computing is growing rapidly is the various benefits it offers.

□ **Reduced costs:** Maintaining IT systems requires big outlays of capital, something that cloud helps reduce. By using the resources provided by the cloud provider, businesses avoid the need to purchase expensive infrastructure, substantially reducing their expenditure. Cloud providers work on the pay-as-you-go model, which means businesses only pay for the services they use, further reducing costs.

□ **Scalability:** Cloud allows organizations to grow their users from merely a few to thousands in a very short time. Depending on the need, a business can scale their storage needs up or down, allowing organizations to be fluid.

□ **Flexibility and collaboration:** Since the data on cloud can be accessed directly via the internet, it gives employees the ability to work from anywhere, anytime. Cloud gives you the freedom to set up your virtual office anywhere you are. It also allows teams to work on a project across locations by giving them access to the same files as third-party vendors.

□ **Business continuity:** Cloud safely stores and protects your data in the event of an outage or crisis. This makes it easier to resume work once the systems are up and running again.

□ **Competitive edge:** Cloud takes care of various business aspects, such as maintaining the IT infrastructure, licensing software, or training personnel to manage your data. It, therefore, gives you an edge over your competitors since the time and resources you invest are minimal.

CHALLENGES

Now let's move on to discuss some challenges of cloud computing.

Downtime: Almost every cloud user will tell you that outages top their list of cloud computing challenges. At times, cloud service providers may get overwhelmed due to the huge number of clients they provide services to each day. This may lead to technical outages, due to which your applications may temporarily experience some downtime.

Internet connection dependency: A user may not be able to access data on the cloud without a good internet connection and a compatible device. Moreover, using public Wi-

Fi to access your files could pose a threat if the right security measures are not taken.

Financial commitment: Cloud providers use a pay-as-you-go pricing model. However, businesses need to give a monthly or annual financial commitment for most subscription plans. This needs to be factored into their operating costs.

Limited access: A user may have minimal control since the cloud service provider owns and manages the infrastructure. The user would only be able to manage applications and not the backend infrastructure. Crucial tasks, such as firmware management, may not be passed to the user at all. You always have to trust a third-party vendor to ensure security and take care of your data.

Security risks: Even if your cloud service provider promises you that they have the most reliable security certifications, there's always a chance of losing your data. With hackers increasingly targeting cloud storage to gain access to sensitive business data, this might be an even greater concern, for which the appropriate measures need to be taken.

Lecture three

Distributed

Computing

A distributed computer system consists of multiple software components that are on multiple computers, but run as a single system. The computers that are in a distributed system can be physically close together and connected by a local network, or they can be geographically distant and connected by a wide area network.

Definition: Distributed computing is a model in which components of a software system are shared among multiple computers or nodes. Even though the software components may be spread out across multiple computers in multiple locations, they are run as one system. This is done to improve efficiency and performance. The systems on different networked computers communicate and coordinate by sending messages back and forth to achieve a defined task.

A distributed system can consist of any number of possible configurations, such as mainframes, personal computers, workstations, minicomputers, and so on. The goal of distributed computing is to make such a network work as a single computer. Common use cases of a distributed systems are electronic banking systems, massive multiplayer online games, and sensor networks.

HOW DOES DISTRIBUTED COMPUTING WORK?

Distributed computing works by computers passing messages to each other within the distributed systems architecture.

- Each component of the system works to achieve a common goal and the end-user views results as one combined unit.
- Each component has its own end-user and the distributed system facilitates sharing resources or communication services.
- Communication protocols or rules create a dependency between the components of the distributed system. This interdependence is called coupling, and there are two main types of coupling:

1. Loose coupling – In loose coupling, components are weakly connected so that changes to one component do not affect the other (i.e., the components can function independently).
2. Tight coupling – In tight coupling, two or more components that interact with one another are dependent and must be present and functioning at the same time (i.e., every component is dependent on the other).

WHAT ARE THE TYPES OF DISTRIBUTED COMPUTING ARCHITECTURE?

In distributed computing, you design applications that can run on several computers instead of on just one computer. You achieve this by designing the software so that different computers perform different functions and communicate to develop the final solution.

There are four main types of distributed architecture:

1. Client-server architecture – Clients make requests to the servers for data, then formats it and displays it to the end-user. Servers manage access to resources and respond to client requests with data or status information.
2. Three-tier architecture – In three-tier distributed systems, client machines remain as the first tier you access. Server machines, on the other hand, are further divided into two categories:
 - a. Application servers act as the middle tier for communication. They contain the application logic or the core functions that you design the distributed system for.
 - b. Database servers act as the third tier to store and manage the data. They are responsible for data retrieval and data consistency.
3. N-tier architecture – N-tier models include several different client-server systems communicating with each other to solve the same problem. Most modern distributed systems use an n-tier architecture with different enterprise applications working together as one system behind the scenes.
4. Peer-to-peer architecture – Peer-to-peer distributed systems assign equal responsibilities to all networked computers. There is no separation between client and server computers, and any computer can perform all responsibilities. Peer-to-peer architecture has become popular for content sharing, file streaming, and blockchain networks.

CHARACTERISTICS OF DISTRIBUTED SYSTEMS

Computers in distributed systems may be on separate continents, in the same building, or the same room. However, they share the following:

- Heterogeneity. Usually, different platforms working together as one.
- Independent Failures. Each computer can fail independently of the others and independent from the communication channel.
- Parallel activities. Autonomous components executing concurrent tasks.
- Communication via message passing. No shared memory.
- Resource sharing. Printer, database, other services.

- No global state. No single process can have knowledge of the current global state of the system.
- No global clock. Only limited precision for processes to synchronize their clocks.

WHAT ARE THE ADVANTAGES OF DISTRIBUTED COMPUTING?

1. Scalability – Distributed computing systems can grow with your workload and requirements. You can add more computing devices (new nodes), to the distributed computing network when they are needed.
2. Consistency – Computers in a distributed system share information and duplicate data between them, but the system automatically manages data consistency across all the different computers.
3. Transparency – Distributed systems provide logical separation between the user and the physical devices. You can have different hardware, middleware, and software systems working together without worrying about the setup and configuration of individual machines.
4. Performance – Distributed computing can help improve performance by having each computer in a cluster handle different parts of a task simultaneously.
5. Resilience and Redundancy – Multiple computers can provide the same services. This way, if one machine isn't available, others can fill in for the service.
6. Cost-effectiveness – Distributed computing can use low-cost, off-the-shelf hardware.
7. Efficiency – Complex requests can be broken down into smaller pieces and distributed among different systems. This way, the request is simplified and worked on as a form of parallel computing, reducing the time needed to compute requests.
8. Distributed Applications – Unlike traditional applications that run on a single system, distributed applications run on multiple systems simultaneously.

WHAT ARE SOME DISTRIBUTED COMPUTING USE CASES?

Distributed computing is everywhere today. Mobile and web applications are examples of distributed computing because several machines work together in the backend for the application to give you the correct information. However, when distributed systems are scaled up, they can solve more complex challenges.

- Healthcare and life sciences use distributed computing to model and simulate complex life science data. Image analysis, medical drug research, and gene structure analysis all become faster with distributed systems.
- In Engineering research, engineers can simulate complex physics and mechanics concepts on distributed systems. They use this research to improve product design, build complex structures, and design faster vehicles.
- Financial services firms use distributed systems to perform high-speed economic simulations that assess portfolio risks, predict market movements, and support financial decision-making.
- Energy companies need to analyse large volumes of data to improve operations and

transition to sustainable and climate-friendly solutions. They use distributed systems to analyse high-volume data streams from a vast network of sensors and other intelligent devices.

CLOUD COMPUTING VS. DISTRIBUTED COMPUTING

Cloud and distributed computing both focus on spreading a service or services to a number of different machines; however, cloud computing typically offers a service like a specific software or storage for organizations to use on their own tasks. Meanwhile, distributed computing involves distributing services to different computers to aid in or around the same task.

Distributed computing alongside virtualization, Web 2.0 technologies, and service orientation makes up the core technologies that enables the provision of cloud services from anywhere on the globe.

- Distributed computing is a foundational model for cloud computing because cloud systems are distributed systems.

- Virtualization is a core feature of the infrastructure used by cloud providers, which plays a fundamental role in cloud computing.

- Web 2.0 technologies constitute the interface through which cloud computing services are delivered, managed, and provisioned. Besides the interaction with rich interfaces through the Web browser, Web services have become the primary access point to cloud computing systems.

- Service orientation is the underlying paradigm that defines the architecture of a cloud computing system (SaaS, PaaS, IaaS). Outside these categories, there are other examples such as Anything as a service (XaaS), disaster recovery as a service (DRaaS), communications as a service (CaaS), network as a service (NaaS), database as a service (DBaaS), storage as a service (STaaS), desktop as a service (DaaS), and monitoring as a service (MaaS).

PARALLEL COMPUTING VS. DISTRIBUTED COMPUTING

- Parallel computing is a type of computing in which one computer or multiple computers in a network carry out many calculations or processes simultaneously.

- Parallel computing is a particularly tightly coupled form of distributed computing. In parallel processing, all processors have access to shared memory for exchanging information between them. On the other hand, in distributed processing, each processor has private memory (distributed memory). Processors use message passing to exchange information.

GRID COMPUTING VS. DISTRIBUTED COMPUTING

- Grid computing is a computing model involving a distributed architecture of multiple computers connected to solve a complex problem.

- In the grid computing model, servers or PCs run independent tasks and are linked loosely by the internet or low-speed networks.

- Grid computing is highly scaled distributed computing that emphasizes performance and coordination between several networks. Internally, each grid acts like a tightly coupled

computing system. However, externally, grids are more loosely coupled.

□ Each grid network performs individual functions and communicates the results to other grids.

Lecture four

Mobile & Wireless

computing

Mobile Computing is a technology that allows transmission of data, voice and video via a computer or any other wireless enabled device without having to be connected to a fixed physical link.

Definition: Mobile computing refers to the set of IT technologies, products, services and operational strategies and procedures that enable end users to access computation, information and related resources and capabilities while mobile. Mobile most commonly refers to access in motion, where the user is not restricted to a given geographic location.

Today, mobile computing technology is everywhere. It has applications in business and consumer markets, industrial and entertainment industries, as well as many specialized vertical markets.

Desktop computers offer more hardware configuration capabilities and computational performance. However, the majority of end users prefer mobile devices. The key advantage of mobile computing is convenience, where users have access to information and computational resources anytime and anywhere.

HOW DOES MOBILE COMPUTING WORK?

□ Generally, a mobile computing system involves:

- o A mobile device, such as a laptop computer, tablet or smartphone, and
- o A wireless network connection based on Wi-Fi or cellular wireless technology, such as 5G.

□ Mobile devices typically can store data locally, and access to that data doesn't require a network connection.

□ Mobile computers typically accommodate access to both wireless and wired technology. Access to shared network resources, including Mobile cloud-based resources, is essential given the collaborative nature of work today.

□ Integrated, rechargeable batteries power mobile devices, and most can run on an alternating current (AC) power source when used from a fixed location.

□ Mobile device selection is based on the intended application selected for use. For example, laptops are better suited for content creation, and tablets are often preferred for content consumption. Smartphones function as pocket-size computers and communication devices, but they have small screen sizes and screen-based keyboards.

WHY IS MOBILE COMPUTING USED?

Mobile computing is used in most facets of life both in business and by consumers.

□ Users can use mobile devices like laptops, smartphones, wearables and tablets for an array

of activities, including communication, entertainment, banking, bill paying and health and fitness tracking.

□ It enables users to be untethered from a power source for periods of time. This is advantageous for traveling workers who want to stay connected to their work while on the move but may not have all the connectivity and power options they have in an office setting.

□ It's also useful for remote workers. Employees use laptops or tablets to work and collaborate. Delivery personnel use mobile devices for logistical and management information, and to verify that deliveries were made to the right place. Field service technicians use mobile devices for field service management, tracking and support capabilities.

□ Mobile devices and apps can collect user data in various environments and contexts. Fitbits and smartwatches are examples of wearable technology that collect user data in novel contexts, including fitness and health settings.

□ Mobile computing also makes the internet of things (IoT) possible. Consumer and enterprise IoT devices are used to communicate with other devices without human intervention. For example, self-driving cars use sensors, onboard computers and other connected devices to connect with GPS, weather and other systems to navigate the road safely. Smart sensors are also used in supply chain management systems to track the progress of goods in transit.

WHAT ARE THE TYPES OF MOBILE COMPUTING?

Mobile computing is a combination of infrastructure, hardware and software technology. The various parts of mobile computing are as follows:

1. **Mobile Infrastructure.** Infrastructure refers to the technical pieces that allow devices to communicate. Mobile infrastructure includes the wireless networks, wireless protocols and data formats.
2. **Mobile Hardware.** The physical mobile device and supporting hardware that users interact with make up the mobile hardware. This may include cell phones, laptops, tablets, wearable computers and accompanying chargers and accessories.
3. **Mobile Software.** This refers to the applications that run on mobile devices, including mobile operating systems (OSes) and user-facing applications, such as mobile browsers and e-commerce applications.

MOBILE CLASSIFICATION

Mobile computing is not only limited to mobile phones, but there are various gadgets available in the market that are built on a platform to support mobile computing. They are usually classified in the following categories:

□ **Personal Digital Assistant (PDA)**

□ The main purpose of this device is to act as an electronic organizer or day planner that is portable, easy to use and capable

of sharing information with your computer systems.

□ PDA is an extension of the PC, not a replacement. These systems are capable of sharing information with a computer system through a process or service known as synchronization. Both devices will access each other to check for changes or updates in the individual devices. The use of infrared and Bluetooth connections enables these devices to always be synchronized.

□ With PDA devices, a user can browse the internet, listen to audio clips, watch video clips, edit and modify office documents, and many more services. The device has a stylus and a touch sensitive screen for input and output purposes.

□ Smartphones

□ It combines the features of a PDA with that of a mobile phone or camera phone. It has a superior edge over other kinds of mobile phones.

□ Smartphones have the capability to run multiple programs concurrently. These phones include high-resolution touch screens, and web browsers that can access and properly display standard web pages rather than just mobile-optimized sites, and can access high-speed data via Wi-Fi and high-speed cellular broadband.

□ The most common mobile Operating Systems (OS) used by modern smartphones include, Google's Android, Apple's iOS, Nokia's Symbian, RIM's BlackBerry OS, Samsung's Bada, Microsoft's Windows Phone, and embedded Linux distributions such as Maemo and MeeGo.

□ Such operating systems can be installed on different phone models, and typically each device can receive multiple OS software updates over its lifetime.

□ Tablet PC and iPads

□ This mobile device is larger than a mobile phone or a PDA. It integrates into a touch screen and is operated using touch sensitive motions on the screen. They are often controlled by a pen or by the touch of a finger.

□ They are usually in slate form and are light in weight. Examples would include iPads, Galaxy Tabs, Blackberry Playbooks etc.

□ They offer the same functionality as portable computers and they support mobile computing in a far superior way and have enormous processing horsepower.

□ Users can edit and modify document files, access high speed internet, stream video and audio data, receive and send e-mails, attend/give lectures and presentations among its very many other functions. They have excellent screen resolution and clarity

WHAT ARE THE ADVANTAGES OF MOBILE COMPUTING?

1. Flexibility — Mobile computing allows you to work from anywhere, at any time, giving you the flexibility to get things done when and where it's convenient for you.
2. Connectivity — Mobile computing keeps you connected to the Internet, allowing you to access information, stay in touch with others, and collaborate with colleagues and classmates.
3. Efficiency — Mobile computing helps you be more efficient, allowing you to complete tasks quickly and easily while on the go.
4. Productivity — Mobile computing can increase productivity, as you can take care of important tasks and responsibilities while away from your desk.
5. Innovation — Mobile computing is constantly advancing and evolving, providing new and innovative ways to stay connected and get things done.

WHAT ARE THE LIMITATIONS OF MOBILE COMPUTING?

1. Connectivity — While the mobile infrastructure continues to improve, there are areas where signal strength is poor or non-existent.
2. Distraction — Mobile devices can be distracting and potentially dangerous in a hazardous work environment that requires the employee's attention, such as a construction site. They pose dangers if used inappropriately while driving.
3. Dependence — Mobile computing can lead to a dependence on technology, making it difficult to disconnect and enjoy life without constant digital distraction.
4. Battery Life — Mobile devices have limited battery life, which requires frequent recharging, thus causing inconvenience and disruption.
5. Security Concerns — Mobile devices can be vulnerable to cyberattacks and data theft, putting personal information and sensitive data at risk.
6. Cost — Mobile computing can be expensive, with high costs for devices, data plans, and accessories.
7. Physical Damage — Mobile devices are small and fragile, making them susceptible to physical damage from drops, spills, and other accidents.

MOBILE COMPUTING SECURITY THREATS

Mobile computing has its fair share of security concerns as any other technology. The first step to securing mobile devices is knowing the risks that are lurking on the Internet that might be targeted at you. Cybercriminals can be after your data, your credentials, or your money, and they can use multiple ways to get what they want, some of the which includes:-

- Phishing – This is a scamming attempt to steal users' credentials or sensitive data, such as credit card numbers. Fraudsters send users emails or short message service (SMS) messages (commonly known as text messages) designed to look as though they're coming from a legitimate source, using fake hyperlinks.
- Malware and Ransomware – Mobile malware is undetected software, such as a malicious app or spyware, created to damage, disrupt or gain illegitimate access to a client, computer, server or computer network. Ransomware, a form of malware, threatens to destroy or

withhold a victim's data or files unless a ransom is paid to decrypt files and restore access.

□ Cryptojacking – This is a form of malware, uses an organization's computing power or individual's computer power without their knowledge to mine cryptocurrencies such as Bitcoin or Ethereum, decreasing a device's processing abilities and effectiveness.

□ Unsecured Wi-Fi – Hotspots without a virtual private network (VPN) make mobile devices more vulnerable to cyberattack. Cybercriminals can intercept traffic and steal private information using methods such as man-in-the-middle (MitM) attacks. Cybercriminals can also deceive users into connecting to rogue hotspots, making it easier to extract corporate or personal data.

□ Outdated operating systems usually contain vulnerabilities that have been exploited by cybercriminals, and devices with outdated OS remain vulnerable to attack. Manufacturer updates often include critical security patches to address vulnerabilities that may be actively exploited.

□ Excessive app permissions. Mobile apps have the power to compromise data privacy through excessive app permissions. App permissions determine an app's functionality and access to a user's device and features, such as its microphone and camera. Some apps are riskier than others. Some can be compromised, and sensitive data can be funnelled through to untrustworthy third parties.

□ The Internet of Things (IoT). Over the years, the number and type of devices – from phones to smartwatches and wireless appliances – that can access your network has grown so fast, which can be hard to monitor and secure with an antivirus solution. That is why IoT devices can be tempting for hackers who target them as an entry point to your home or business network.

□ Password security. Bad password habits like reusing passwords or choosing weak passwords from the get-go just make a cybercriminal's work much easier. Additionally, you can be at risk for credential-based brute-force cyberattacks like credential stuffing or password spraying.

Lecture five

NETWORK

SECURITY

INTRODUCTION

The transmission of data from one point (A) on the network to the other point (B) is a great concern and therefore, there is the need to deploy measures that can secure the transmission of data away from unauthorized individuals. Hence, the need for network security.

Definition

□ Network security is a term that describes the security tools, tactics and security policies designed to monitor, prevent and respond to unauthorized network intrusion, while also protecting digital assets, including network traffic.

□ Network security includes hardware and software technologies (including resources such

as savvy security analysts, hunters, and incident responders) and is designed to respond to the full range of potential threats targeting your network.

□ Network security is the protection of the underlying networking infrastructure from unauthorized access, misuse, or theft. It involves creating a secure infrastructure for devices, applications, users, and applications to work in a secure manner.

Data as the Life-Blood of Businesses

Data is the lifeblood of any business. It supports business growth, carries vital resources and helps organizations to stay healthy. And if data is blood, then the network is the heart beat that pumps data through the business system. But modern cyber threats are like vampires, doing everything possible to get that blood that keeps any business going. In order to defend against these vampiric threats and save your business from data loss that could potentially be worth millions, you need more than just a stake and some cloves of garlic; you need effective, robust network security and network visibility.

How does network security work?

Network security combines multiple layers of defenses at the edge and in the network. Each network security layer implements policies and controls. Authorized users gain access to network resources, but malicious actors are blocked from carrying out exploits and threats.

Key Focuses of Network Security

There are three key focuses that should serve as a foundation of any network security strategy: protection, detection and response.

- Protection entails any tools or policies designed to prevent network security intrusion.
- Detection refers to the resources that allow you to analyse network traffic and quickly identify problems before they can do harm.
- Response is the ability to react to identified network security threats and resolve them as quickly as possible.

Network Security Tools and Techniques

Enterprises' network face threats of all shapes and sizes, and thus should be prepared to defend, identify and respond to a full range of attacks. But the reality is that the biggest danger to most companies are not fly-by-night threat actors, but rather attackers that are well-funded and are targeting specific organizations for specific reasons. For that reason, your network security strategy needs to be able to address the various methods these actors might employ.

Here are some network security tools and techniques designed to help you do just that:

Access control

If threat actors cannot access your network, the amount of damage they will be able to do will be extremely limited. But in addition to preventing unauthorized access, be aware that even authorized users can also be potential threats. Access control allows you to increase your network security by limiting user access and resources to only the parts of the network that directly apply to individual users' responsibilities.

Anti-malware software

Malware, in the form of viruses, trojans, worms, keyloggers, spyware, etc. are designed to spread

through computer systems and infect networks. Anti-malware tools are a kind of network security software designed to identify dangerous programs and prevent them from spreading. Anti-malware and antivirus software may also be able to help resolve malware infections, minimizing the damage to the network.

Anomaly detection

It can be difficult to identify anomalies in your network without a baseline understanding of how that network should be operating. Network anomaly detection engines (ADE) allow you to analyse your network, so that when breaches occur, you'll be alerted to them quickly enough to be able to respond.

Application security

For many attackers, applications are a defensive vulnerability that can be exploited. Application security helps establish security parameters for any applications that may be relevant to your network security.

Data loss prevention (DLP)

Often, the weakest link in network security is the human element. DLP technologies and policies help protect staff and other users from misusing and possibly compromising sensitive data or allowing said data out of the network.

Email security

As with DLP, email security is focused on shoring up human-related security weaknesses. Via phishing strategies (which are often very complex and convincing), attackers persuade email recipients to share sensitive information via desktop or mobile device, or inadvertently download malware into the targeted network. Email security helps identify dangerous emails and can also be used to block attacks and prevent the sharing of vital data.

Endpoint security

The business world is becoming increasingly bring your own device (BYOD), to the point where the distinction between personal and business computer devices is almost non-existent. Unfortunately, sometimes the personal devices become targets when users rely on them to access business networks. Endpoint security adds a layer of defense between remote devices and business networks.

Firewall

Firewall function much like gates that can be used to secure the borders between your network and the internet. Firewalls are used to manage network traffic, allowing authorized traffic through while blocking access to non-authorized traffic.

Intrusion prevention systems

Intrusion prevention systems (also called intrusion detection) constantly scan and analyse network traffic/packets, so that different types of attacks can be identified and responded to quickly. These systems often keep a database of known attack methods, so as to be able to recognize threats immediately.

Network segmentation

There are many kinds of network traffic, each associated with different security risks. Network

segmentation allows you to grant the right access to the right traffic, while restricting traffic from suspicious sources.

Security information and event management (SIEM)

Sometimes simply pulling together the right information from so many different tools and resources can be prohibitively difficult — particularly when time is an issue. SIEM tools and software give responders the data they need to act quickly.

Virtual private network (VPN)

VPN tools are used to authenticate communication between secure networks and an endpoint device. Remote-access VPNs generally use IPsec or Secure Sockets Layer (SSL) for authentication, creating an encrypted line to block other parties from eavesdropping.

Remote Access VPN

Remote access VPN provides remote and secure access to a company network to individual hosts or clients, such as telecommuters, mobile users, and extranet consumers. Each host typically has VPN client software loaded or uses a web-based client. Privacy and integrity of sensitive information is ensured through multi-factor authentication, endpoint compliance scanning, and encryption of all transmitted data.

Zero Trust Network Access (ZTNA)

The zero trust security model states that a user should only have the access and permissions that they require to fulfil their role. This is a very different approach from that provided by traditional security solutions, like VPNs, that grant a user full access to the target network. Zero trust network access (ZTNA) also known as software-defined perimeter (SDP) solutions permits granular access to an organization's applications from users who require that access to perform their duties.

Web security

Including tools, hardware, policies and more, web security is a blanket term to describe the network security measures businesses take to ensure safe web use when connected to an internal network. This helps prevent web-based threats from using browsers as access points to get into the network.

Wireless security

Generally speaking, wireless networks are less secure than traditional networks. Thus, strict wireless security measures are necessary to ensure that threat actors aren't gaining access.

Sandboxing

Sandboxing is a cybersecurity practice where you run code or open files in a safe, isolated environment on a host machine that mimics end-user operating environments. Sandboxing observes the files or code as they are opened and looks for malicious behavior to prevent threats from getting on the network. For example malware in files such as PDF, Microsoft Word, Excel and PowerPoint can be safely detected and blocked before the files reach an unsuspecting end user.

Hyperscale Network Security

Hyperscale is the ability of an architecture to scale appropriately, as increased demand is added

to the system. This solution includes rapid deployment and scaling up or down to meet changes in network security demands. By tightly integrating networking and compute resources in a software-defined system, it is possible to fully utilize all hardware resources available in a clustering solution.

Cloud Network Security

Applications and workloads are no longer exclusively hosted on-premises in a local data center. Protecting the modern data centre requires greater flexibility and innovation to keep pace with the migration of application workloads to the cloud. Software-defined Networking (SDN) and Software-defined Wide Area Network (SD-WAN) solutions enable network security solutions in private, public, hybrid and cloud-hosted Firewall-as-a-Service (FWaaS) deployments.

ROBUST NETWORK SECURITY WILL PROTECT AGAINST

Virus

A virus is a malicious, downloadable file that can lay dormant that replicates itself by changing other computer programs with its own code. Once it spreads those files are infected and can spread from one computer to another, and/or corrupt or destroy network data.

Worms

Can slow down computer networks by eating up bandwidth as well as the slow the efficiency of your computer to process data. A worm is a standalone malware that can propagate and work independently of other files, where a virus needs a host program to spread.

Trojan

A trojan is a backdoor program that creates an entryway for malicious users to access the computer system by using what looks like a real program, but quickly turns out to be harmful. A trojan virus can delete files, activate other malware hidden on your computer network, such as a virus and steal valuable data.

Spyware

Much like its name, spyware is a computer virus that gathers information about a person or organization without their express knowledge and may send the information gathered to a third party without the consumer's consent.

Adware

Can redirect your search requests to advertising websites and collect marketing data about you in the process so that customized advertisements will be displayed based on your search and buying history.

Ransomware

This is a type of trojan cyberware that is designed to gain money from the person or organization's computer on which it is installed by encrypting data so that it is unusable, blocking access to the user's system.

Benefits of Network Security

Network Security Builds trust

Security for large systems translates to security for everyone. Network security boosts client and consumer confidence, and it protects your business from the reputational and legal fallout of a security breach.

Network Security Mitigates risk

The right network security solution will help your business stay compliant with business and government regulations, and it will minimize the business and financial impact of a breach if it does occur.

Network Security Protects proprietary information

Your clients and customers rely on you to protect their sensitive information. Your business relies on that same protection, too. Network security ensures the protection of information and data shared across the network.

Network Security Enables a more modern workplace

From allowing employees to work securely from any location using VPN to encouraging collaboration with secure network access, network security provides options to enable the future of work. Effective network security also provides many levels of security to scale with your growing business.

Client/Server

Computing

(using the web)

In client-server computing, the clients request a resource and the server provides that resource. A server may serve multiple clients at the same time while a client is in contact with only one server. Both the client and server usually communicate via a computer network but sometimes they may reside in the same system.

Characteristics of Client Server Computing

The salient points for client server computing are as follows:

- The client server computing works with a system of request and response. The client sends a request to the server and the server responds with the desired information.
- The client and server should follow a common communication protocol so they can easily interact with each other. All the communication protocols are available at the application layer.
- A server can only accommodate a limited number of client requests at a time. So, it uses a system based to priority to respond to the requests.
- Denial of Service (DoS) attacks hinder a servers' ability to respond to authentic client requests by inundating it with false requests.
- An example of a client server computing system is a web server. It returns the web pages to the clients that requested them.

Difference between Client Server and Peer-to-Peer Computing

The major differences between client server computing and peer to peer computing are as follows:

□ In client server computing, a server is a central node that services many client nodes. On the other hand, in a peer-to-peer system, the nodes collectively use their resources and communicate with each other.

□ In client server computing the server is the one that communicates with the other nodes. In peer to peer to computing, all the nodes are equal and share data with each other directly.

□ Client Server computing is believed to be a subcategory of the peer to peer computing.

Advantages of Client Server Computing

The different advantages of client server computing are –

□ All the required data is concentrated in a single place i.e., the server. So, it is easy to protect the data and provide authorisation and authentication.

□ The server need not be located physically close to the clients. Yet the data can be accessed efficiently.

□ It is easy to replace, upgrade or relocate the nodes in the client server model because all the nodes are independent and request data only from the server.

□ All the nodes (clients and server) may not be built on similar platforms yet they can easily facilitate the transfer of data.

Disadvantages of Client Server Computing

The different disadvantages of client server computing are –

□ If all the clients simultaneously request data from the server, it may get overloaded. This may lead to congestion in the network.

□ If the server fails for any reason, then none of the requests of the clients can be fulfilled. This leads of failure of the client server network.

□ The cost of setting and maintaining a client server model are quite high.

Building

Web

Applications

A Web app is an interactive computer program, built with web technologies (HTML, CSS, JS), which stores (Database, Files) and manipulates data (CRUD), and is used by a team or single user to perform tasks over the internet. The HTML and the CSS serves as the front-end to receive data from the user while the database, programming like JavaScript and PHP serves as the back-end.

Prerequisites for Building a Web Application

To make a data-centric web app from the bottom-up, it is advantageous to understand:

1. Backend language (e.g., Python, Ruby) - control how your web app works
2. Web front end (HTML, CSS, JavaScript) - for the look and feel of your web app
3. DevOps (GitHub, Jenkins) - Deploying / hosting your web app

If you do not have any experience with the points above, you need not worry. You have two options:

□ Learn the points above. There are lots of resources online to help you. I'd recommend:

- o FreeCodeCamp
- o Codecademy
- o Codewars

☐ Use a web app builder. Web app builders allows you to build web apps without writing any code. I'd recommend:

- o Google AppSheet
- o Bubble
- o Glide

Building a Web Application

Step 1 – Source an Idea

Before making a web app, you must first understand what you intend on building, and more importantly, the reason for it. Your idea should stem from solving someone's problem. Ideally, it is important that you choose an idea which is of your interest. This can be done by asking ey questions like:

- ☐ How much time do I have to build this app?
- ☐ What am I interested in?
- ☐ What apps do I enjoy using?
- ☐ What do I like about these apps?
- ☐ How much time/money will this app save or generate for me (as a user)?
- ☐ How much will it improve my life?

Step 2 – Market Research

Once you have chosen your idea(s), it is important to research the market to see if a similar product exists, and if a market exists. The number major reason why start-ups fail is due to the failure of to achieving product-market fit. Product/market fit means being in a good market with a product that can satisfy that market. To quickly find out if a similar web app exists, use the following tools to search for your idea:

- ☐ Google
- ☐ Patent and trademark search
- ☐ Betalist

If a similar product exists, do not worry. This can be a sign that a market for your new idea exists. Your future competitors have laid the groundwork, educated the market. It's time for you to swoop in and steal the thunder.

If a similar product does not exist, it is a possibility you have struck lucky. On the other hand, it is a possibility someone before has ventured down this path and hit a dead-end. Nobody wants to experience that, so it is important to dive deep into the market and source the wisdom of:

1. Your Web App's target market - Share your web app idea on forums related to your target market. If you know anyone who works within your target market, explain your idea to them. The more you talk and receive validation from your target market, the better.
2. Google Trends - A quick search of your web app idea will reveal relating trends.
3. SEO tool - Write a list of keywords relating to your web app. If it's an 'OKR tool', use the

tools to search 'OKR tool', 'OKR app', and 'objectives and key results software'. If the SEO tool indicates there are lots of people searching for your keyword terms, this is a small indicator you have a target market.

4. Social Media - Jump over to Twitter/Facebook groups and present your idea to your target market.

5. Events - If there is a local event in your area attracting people from your target market, go to it. Share your idea and record the feedback. After completing the above steps, you should have enough information to understand if there's a market for your product. If there is a market for your product, and there's also established competition, it's important to research them.

Step 3 - Define your web apps functionality

You have got your idea, you have validated the market, it is now time to list everything you want your app to do. A common mistake here is to get carried away. The more functionality you add, the longer it will take to build your web app. Quite often, the longer a web app takes to build, the more frustration you will experience.

Only define functionality which solves your target markets problems. Remember, your web app is a work in progress and the first goal is version 1. It will still have cool features and delight your users, but you must keep things simple. For direction, below is a list of basic functions required for a simple Customer Relationship Management (CRM) app:

- ☐ Users can create an account
- ☐ Users can retrieve lost passwords
- ☐ Users can change their passwords
- ☐ Users can create new contacts
- ☐ Users can upload new contacts
- ☐ Users can assign a value to contacts
- ☐ Users can write notes under contacts
- ☐ Users can label a contact as a lead, customer, or associate
- ☐ Users can filter contacts by lead, customer, or associate
- ☐ Users can view the total value of leads, customers and associates

The above list will help you define your features. Once you're done, roll up your sleeves. It's time to get creative! Moving from the Ideation stage, to design stage.

Step 4 - Sketch your web app

There are multiple stages of designing a web app. The first stage is sketching using a notebook (with no lines) and pen/pencil.

After step 1, 2 and 3, you should have an idea of what your web app is, who your users are, and the features it will have. Sketch out the wireframe of your web apps UI (User Interface). It does not have to be exact; it is just a sketch. When sketching, consider the following:

- ☐ Navigation
- ☐ Branding
- ☐ Forms

- Buttons
- Any other interactive elements

Sketch different versions of your web app. Consider how your web app's functionality will affect the overall design. Annotate your sketch and outline how your app should work. Taking notes will help you clarify and understand why you have designed certain elements at a later stage. Overcomplicating the design at this stage will only lead to frustration.

Step 5 – Plan your web apps workflow

It is time to put yourself in the shoes of your user. Here, you plan your web apps workflow. First, go back to step 2 and look at your market research. Take your list of competitors and sign up to their free trials. Have a quick play around with their product. Write notes on what you thought was good and what you thought was bad. Pay particular attention to the workflow.

After you have finished analysing your competitor's web apps, it is time to write down different workflows for your app. Consider the following points:

- How does a user signup?
- Do they receive a verification email?
- How does a user login?
- How does a user change their password?
- How does a user navigate through the app?
- How does a user change their user settings?
- How does a user pay for the app?
- How does a user cancel their subscription?

All of a sudden our one-page web app turns into a 10-page web app. Write a list of all the different pages your web application will have. Consider the different states of pages. For example, the homepage will have two states; logged in and logged out. Logged in users will see a different page than logged out users, and so on.

Step 6 – Wireframing / Prototyping Your Web Application

Ok, it's time to turn those sketches and that new-found understanding of your web application into a wireframe/prototype. Wireframing is the process of designing a blueprint of your web application. Prototyping is taking wireframing a step further, adding an interactive display.

The decision is to wireframe or prototype is up to you. If you have the time, I would have recommended prototyping as it will make it easier to communicate your web app when seeking validation. You can prototype/wireframe using the following tools:

Sketch (macOS)

InVision (macOS)

Adobe XD (macOS, Windows)

Figma (Web, macOS, Windows, Linux)

Balsamiq (macOS, Windows, Web)

I recommend you create a design system/style guide first. You can find inspiration at UXPIn. Design systems improve design consistency. But it's not required.

Step 7 – Seek early validation

You have now got a beautiful wireframe/prototype which visually describes your web app. It is time to show your beautiful wireframe to the world. At this stage we want constructive feedback.

Simply asking your friends would they use your new web app is not enough. You should start with a small number of representative users. Go to your target market's forums, watering holes, their places of work and verify the problem with them, and present your solution. Try to build a rapport with these representatives as they could become your customers. Once you have gathered feedback and product validation about your prototype, you can start building your web app.

However, before Starting the development stage, take note of the following tips:

- Attempt to get a small section of your app fully working. Building the smallest possible section will allow you to piece all the bits together, and iron out those creases early.
- At the start - expect things to change a lot as you learn and discover what you have not thought about. Have faith that your app will stabilise and don't be afraid of making big changes.
- Spend time learning your tools. You may feel like you are wasting your time, reading, or experimenting with "hello world". However, learning the correct way to do things will have a huge positive, cumulative effect on your productivity over time.
- Do not avoid issues that need to be fixed. Face your issues head on - they will never go away and will only grow in stature. However, If changes are still expected while building your web app, then its best to spend as little time as possible on the issues (it is a tricky balance).

Step 8 – Architect and build your database

So far, we know roughly our web application's functionality, what it looks like, and the pages required. Now it is time to determine what information we will store in our database.

A database is simply a collection of data! Data can be stored to disk, or in memory on a server, or both. You could create a folder on your hard drive, store a few documents, and call it a database.

A Database Management System (DBMS) is a system that provides you with consistent APIs to (most commonly):

- Create databases, update and delete databases,
- Read and write data to databases,
- Secure access to a database by providing levelled access to different areas and functions.

The type of data you need to store and what your users will use the data to do, will determine the type of database required to run your web app.

There are many types of database system for many different purposes. A web app will most commonly use one of the following:

1. SQL – You should use a SQL database if your data is very relational. Your data is relational if you have multiple, well defined record types that have relationships between them. For example, a "Customer" may have many "Invoices" stored against their record. Typically, you would create a Customer table and an Invoice table - which could be linked together by a "Foreign Key" columns. E.g., Customer.Id = Invoice.CustomerId.

SQL databases have an extremely powerful query language that allows you to present your data in all sorts of useful ways. They have been around for decades, are very well understood, and usually a safe choice. MySQL, Postgresql, Microsoft SQLServer are some of the most common, along with many more modern offerings. The downside of SQL databases is that you must declare all your tables and columns up front.

2. Document Database – You should use a document database if your data is not very relational. Document databases store “documents”. Each record in your database is simply a big blob of structured data – often in JSON format. If you need to store relationships between your records, you will have to write code to manage this yourself. However, many other aspects of using document databases are much simpler. Your database can be “schemaless” - meaning that you do not have to declare your records’ definitions up front.

Generally speaking, document database tends to be much more scalable than SQL databases. They usually offer some querying capabilities, although sometimes not as powerful as SQL. Examples of document databases are: MongoDB, CouchDb, Firebase (serverless), Dynamo Db (AWS).

Step 9 - Build the frontend

Note: In reality, you will build your backend and frontend at the same time. But we’ll keep it simple for better understanding.

The Frontend is the visual element of your web application. It defines what you see and interact with. The frontend is developed with HTML, CSS, and JavaScript.

First, you need to set up your development environment. The components of this will be:

- A code editor, such as VS Code, Sublime Text
- A compilation, and packaging framework:
 - o Webpack,
 - o Gulp,
 - o Grunt.
- A frontend framework (strictly not necessary, but highly advised unless you are an experienced frontend developer):
 - o React,
 - o Ember,
 - o Vue,
 - o Svelte, (The list is endless!).
- Configuring your packaging tool to talk to your backend - which is most likely running on a different port on localhost. Usually, this is done using Node’s HTTP proxy. Most packaging solutions have this option built-in, or available as plugins.

Step 10 - Build your backend

The backend is typically what manages your data. This refers to databases, servers, and everything the user can’t see within a web application. Building your backend is one of the toughest parts of web app development.

The primary job of the backend is to:

1. Provide HTTP endpoints for your frontend, which allow it to operate on your data. E.g., Create, Read, Update and Delete (“CRUD”) records.
2. Authenticate users (verify they are who they say they are before logging them in).
3. Authorization. When a logged in user makes a request, the backend will determine whether they are allowed (authorized) to perform the requested action.
4. Serve the frontend.

When building your web app, you need to choose between Single Page Application and Multiple Page Application (Server Pages). Your choice will affect how you develop your backend.

□ If you have chosen Server Pages, your backend will also be responsible for generating your frontend and serving it to your users. Server pages need their own framework:

Django

Express

Flask

□ With a single page app, the backend will simply serve your static frontend files (i.e., your “Single Page” and its related assets).

When choosing your backend, go with what’s familiar.

□ Login/User & Session Management. How will users authenticate?

o Username and password?

o Open ID (i.e., sign in as Google, FB, etc)

□ What is the security best practices employed in your web app?

I highly recommend OWASP.

□ What user levels will you create in the system?

□ Creating multiple environments is highly recommended. For example:

o Testing environment is need for all the latest development features.

o Beta environment is needed to give early releases to clients.

o Production environment is needed for your live web app system.

Step 11 - Host your web application

Hosting involves running your web app on a particular server. To host your web app, follow these quick steps:

1. Buy a domain (e.g., Namecheap).
2. Buy/Setup an SSL certificate (for encryption).
3. Choose a cloud provider:

Amazon, Vercel,

MS Azure, Heroku,

Google Cloud Platform, Firebase, etc..

DigitalOcean,

Alternatively, the above steps can be automated with Budibase (although, you are still required to buy a domain).

Choosing one of these hosting options will almost certainly provide you with everything you

need. They have ample documentation and community support, and are generally reliable options.

Step 12 - Deploy your web app

You have sourced your idea, validated it, designed and developed your web app, and chosen your hosting provider. You're now at the last step.

The deployment step includes how your web application gets from your source control on your computer to your cloud hosting from step 11.

The following development tools provide continuous integration and will help you with deploying your web app to your cloud hosting:

1. GitLab
2. Bitbucket
3. Jenkins

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Lecture four

Mobile & Wireless computing

Mobile Computing is a technology that allows transmission of data, voice and video via a computer or any other wireless enabled device without having to be connected to a fixed physical link.

Definition: Mobile computing refers to the set of IT technologies, products, services and operational strategies and procedures that enable end users to access computation, information and related resources and capabilities while mobile. Mobile most commonly refers to access in motion, where the user is not restricted to a given geographic location.

Today, mobile computing technology is everywhere. It has applications in business and consumer markets, industrial and entertainment industries, as well as many specialized vertical markets.

Desktop computers offer more hardware configuration capabilities and computational performance. However, the majority of end users prefer mobile devices. The key advantage of mobile computing is convenience, where users have access to information and computational resources anytime and anywhere.

HOW DOES MOBILE COMPUTING WORK?

□ Generally, a mobile computing system involves:

- o A mobile device, such as a laptop computer, tablet or smartphone, and
- o A wireless network connection based on Wi-Fi or cellular wireless technology, such as 5G.

□ Mobile devices typically can store data locally, and access to that data doesn't require a network connection.

□ Mobile computers typically accommodate access to both wireless and wired technology. Access to shared network resources, including Mobile cloud-based resources, is essential given the collaborative nature of work today.

□ Integrated, rechargeable batteries power mobile devices, and most can run on an alternating current (AC) power source when used from a fixed location.

□ Mobile device selection is based on the intended application selected for use. For example, laptops are better suited for content creation, and tablets are often preferred for content consumption. Smartphones function as pocket-size computers and communication devices, but they have small screen sizes and screen-based keyboards.

WHY IS MOBILE COMPUTING USED?

Mobile computing is used in most facets of life both in business and by consumers.

□ Users can use mobile devices like laptops, smartphones, wearables and tablets for an array of activities, including communication, entertainment, banking, bill paying and health and

fitness tracking.

□ It enables users to be untethered from a power source for periods of time. This is advantageous for traveling workers who want to stay connected to their work while on the move but may not have all the connectivity and power options they have in an office setting.

□ It's also useful for remote workers. Employees use laptops or tablets to work and collaborate. Delivery personnel use mobile devices for logistical and management information, and to verify that deliveries were made to the right place. Field service technicians use mobile devices for field service management, tracking and support capabilities.

□ Mobile devices and apps can collect user data in various environments and contexts. Fitbits and smartwatches are examples of wearable technology that collect user data in novel contexts, including fitness and health settings.

□ Mobile computing also makes the internet of things (IoT) possible. Consumer and enterprise IoT devices are used to communicate with other devices without human intervention. For example, self-driving cars use sensors, onboard computers and other connected devices to connect with GPS, weather and other systems to navigate the road safely. Smart sensors are also used in supply chain management systems to track the progress of goods in transit.

WHAT ARE THE TYPES OF MOBILE COMPUTING?

Mobile computing is a combination of infrastructure, hardware and software technology. The various parts of mobile computing are as follows:

1. Mobile Infrastructure. Infrastructure refers to the technical pieces that allow devices to communicate. Mobile infrastructure includes the wireless networks, wireless protocols and data formats.
2. Mobile Hardware. The physical mobile device and supporting hardware that users interact with make up the mobile hardware. This may include cell phones, laptops, tablets, wearable computers and accompanying chargers and accessories.
3. Mobile Software. This refers to the applications that run on mobile devices, including mobile operating systems (OSes) and user-facing applications, such as mobile browsers and e-commerce applications.

MOBILE CLASSIFICATION

Mobile computing is not only limited to mobile phones, but there are various gadgets available in the market that are built on a platform to support mobile computing. They are usually classified in the following categories:

□ Personal Digital Assistant (PDA)

□ The main purpose of this device is to act as an electronic organizer or day planner that is portable, easy to use and capable of sharing information with your computer systems.

□ PDA is an extension of the PC, not a replacement. These systems are capable of sharing information with a computer system through a process or service known as synchronization. Both devices will access each other to check for changes or updates in the individual devices. The use of infrared and Bluetooth connections enables these devices to always be synchronized.

□ With PDA devices, a user can browse the internet, listen to audio clips, watch video clips, edit and modify office documents, and many more services. The device has a stylus and a touch sensitive screen for input and output purposes.

□ Smartphones

□ It combines the features of a PDA with that of a mobile phone or camera phone. It has a superior edge over other kinds of mobile phones.

□ Smartphones have the capability to run multiple programs concurrently. These phones include high-resolution touch screens, and web browsers that can access and properly display standard web pages rather than just mobile-optimized sites, and can access high-speed data via Wi-Fi and high-speed cellular broadband.

□ The most common mobile Operating Systems (OS) used by modern smartphones include, Google's Android, Apple's iOS, Nokia's Symbian, RIM's BlackBerry OS, Samsung's Bada, Microsoft's Windows Phone, and embedded Linux distributions such as Maemo and MeeGo.

□ Such operating systems can be installed on different phone models, and typically each device can receive multiple OS software updates over its lifetime.

□ Tablet PC and iPads

□ This mobile device is larger than a mobile phone or a PDA. It integrates into a touch screen and is operated using touch sensitive motions on the screen. They are often controlled by a pen or by the touch of a finger.

□ They are usually in slate form and are light in weight. Examples would include iPads, Galaxy Tabs, Blackberry Playbooks etc.

□ They offer the same functionality as portable computers and they support mobile computing in a far superior way and have enormous processing horsepower.

□ Users can edit and modify document files, access high speed internet, stream video and audio data, receive and send e-mails, attend/give lectures and presentations among its very many other functions. They have excellent screen resolution and clarity

WHAT ARE THE ADVANTAGES OF MOBILE COMPUTING?

1. Flexibility — Mobile computing allows you to work from anywhere, at any time, giving

you the flexibility to get things done when and where it's convenient for you.

2. Connectivity — Mobile computing keeps you connected to the Internet, allowing you to access information, stay in touch with others, and collaborate with colleagues and classmates.

3. Efficiency — Mobile computing helps you be more efficient, allowing you to complete tasks quickly and easily while on the go.

4. Productivity — Mobile computing can increase productivity, as you can take care of important tasks and responsibilities while away from your desk.

5. Innovation — Mobile computing is constantly advancing and evolving, providing new and innovative ways to stay connected and get things done.

WHAT ARE THE LIMITATIONS OF MOBILE COMPUTING?

1. Connectivity — While the mobile infrastructure continues to improve, there are areas where signal strength is poor or non-existent.

2. Distraction — Mobile devices can be distracting and potentially dangerous in a hazardous work environment that requires the employee's attention, such as a construction site. They pose dangers if used inappropriately while driving.

3. Dependence — Mobile computing can lead to a dependence on technology, making it difficult to disconnect and enjoy life without constant digital distraction.

4. Battery Life — Mobile devices have limited battery life, which requires frequent recharging, thus causing inconvenience and disruption.

5. Security Concerns — Mobile devices can be vulnerable to cyberattacks and data theft, putting personal information and sensitive data at risk.

6. Cost — Mobile computing can be expensive, with high costs for devices, data plans, and accessories.

7. Physical Damage — Mobile devices are small and fragile, making them susceptible to physical damage from drops, spills, and other accidents.

MOBILE COMPUTING SECURITY THREATS

Mobile computing has its fair share of security concerns as any other technology. The first step to securing mobile devices is knowing the risks that are lurking on the Internet that might be targeted at you. Cybercriminals can be after your data, your credentials, or your money, and they can use multiple ways to get what they want, some of the which includes:-

□ Phishing – This is a scamming attempt to steal users' credentials or sensitive data, such as credit card numbers. Fraudsters send users emails or short message service (SMS) messages (commonly known as text messages) designed to look as though they're coming from a legitimate source, using fake hyperlinks.

□ Malware and Ransomware – Mobile malware is undetected software, such as a malicious app or spyware, created to damage, disrupt or gain illegitimate access to a client, computer, server or computer network. Ransomware, a form of malware, threatens to destroy or withhold a victim's data or files unless a ransom is paid to decrypt files and restore access.

- Cryptojacking – This is a form of malware, uses an organization’s computing power or individual’s computer power without their knowledge to mine cryptocurrencies such as Bitcoin or Ethereum, decreasing a device’s processing abilities and effectiveness.
- Unsecured Wi-Fi – Hotspots without a virtual private network (VPN) make mobile devices more vulnerable to cyberattack. Cybercriminals can intercept traffic and steal private information using methods such as man-in-the-middle (MitM) attacks. Cybercriminals can also deceive users into connecting to rogue hotspots, making it easier to extract corporate or personal data.
- Outdated operating systems usually contain vulnerabilities that have been exploited by cybercriminals, and devices with outdated OS remain vulnerable to attack. Manufacturer updates often include critical security patches to address vulnerabilities that may be actively exploited.
- Excessive app permissions. Mobile apps have the power to compromise data privacy through excessive app permissions. App permissions determine an app’s functionality and access to a user’s device and features, such as its microphone and camera. Some apps are riskier than others. Some can be compromised, and sensitive data can be funnelled through to untrustworthy third parties.
- The Internet of Things (IoT). Over the years, the number and type of devices – from phones to smartwatches and wireless appliances – that can access your network has grown so fast, which can be hard to monitor and secure with an antivirus solution. That is why IoT devices can be tempting for hackers who target them as an entry point to your home or business network.
- Password security. Bad password habits like reusing passwords or choosing weak passwords from the get-go just make a cybercriminal’s work much easier. Additionally, you can be at risk for credential-based brute-force cyberattacks like credential stuffing or password spraying.